

Halacha

Table of Contents

Concepts	1
Bishul — Cooking	2
מראת עין	3
In בשר וחלב:	4
Cooking:	4
Possible דאורייתא Cooking Methods:	4
Roasting (צלי):	4
Cold into Hot:	4
תתאה גבר:	5
Hot into Cold:	5
Pan-Frying:	6
Smoking (מעושן):	7
Cooking in Hot Springs (חמי טבריא):	7
Electricity:	7
Definite דרבנן Cooking Methods:	8
Soaking (כבוש):	8
Salting (מליחה):	9
Meat and Milk Combined Without Cooking:	9
General Contact:	9
Immersion:	10
Dehydrated Meat:	12
תתאה גבר:	14
Cooking After Removal From the Flame (כלי ראשון/שני/שלישי):	15
כלי ראשון:	15
כלי שני:	15
כלי שלישי:	15
Measurements	16
Depths of Absorption:	16
Surface Level:	16
גרידה:	16
כדי קליפה:	16
כדי נטילה:	17
Marat Ayin — Appearance	18
Applicability to בישול:	19

מראת עין	20
יורה דעה	21
Basar B'Chalav — Meat and Milk	22
The Prohibition:	23
Definitions:	24
Cooking:	24
Possible דאורייתא Cooking Methods:	24
Roasting (צליל):	24
Cold into Hot:	25
תתאה גבר:	26
Hot into Cold:	26
Pan-Frying:	27
Smoking (מעושן):	28
Cooking in Hot Springs (חמי טבריא):	28
Electricity:	28
Definite דרבנן Cooking Methods:	29
Soaking (כבוש):	29
Salting (מליחה):	30
Meat and Milk Combined Without Cooking:	30
General Contact:	30
Immersion:	31
Dehydrated Meat:	33
Meat:	34
Domesticated Animal (בהמה):	34
Blood:	34
Fetus (שליל):	35
Amniotic Sac (שליא):	35
Quasi-Edible Parts:	35
Undomesticated Animal (חיה):	36
Bird (עוף):	36
Eggs:	36
The Development of Chicken Eggs Within the Mother:	37
At What Stage of Development is an Egg Considered an Independent Entity:	37
Fish (דג):	38
Grasshopper (חגב):	38
Milk:	39
דאורייתא:	39
Whey:	39

דרבנן:	39
חיה:	39
מי חלב:	40
חלב מתה:	40
Rennet:	40
Rennet In Cheesemaking:	41
Male Milk:	42
מראת עין:	42
Almond Milk:	42
Mother's Milk:	43
Human Male Milk:	43
Milk of a Non-Kosher Animal:	44
Laws:	45
חתיכה עצמה נעשית נבילה (הענ"נ):	45
Sharp Foods (דבר חריף) and Knives:	45
Sharp Foods (דבר חריף):	45
What's Considered חריף:	45
When Sliced With a Knife:	46
Sharp Foods (דבר חריף):	46
When Sliced With a Knife:	46
If the Status of the Knife is Uncertain:	49
Slicing Non-חריף Foods:	51
Turnip:	51
To Which Prohibitions Does הענ"נ Apply:	52
The Ramifications of הענ"נ:	52
מין במינו and הענ"נ:	53
Localized הענ"נ:	53
Secondary Flavor Transfer (נותן טעם בר נותן טעם):	56
Which Methods of Transfer Effect נב"נ:	58
Sharp Foods (דבר חריף):	60
Washing Dishes:	61
Ashes (Dish Soap):	64
Pouring Hot Water (עירוי):	66
The Status of the Water:	69
Safeguards Against Cooking:	70
The Stringencies of the רמ"א:	70
Safeguards Against Combination:	72
Joint Placement on a Table:	72
What Qualifies as a Reminder:	73
Separate Tablecloths:	73

Bread:	73
A Shared Cup:	73
Waiting Between Meat and Dairy:	74
Dairy After Meat:	74
Length of the Waiting Period:	75
Cleansing the Palate (קינוח):	78
Meat After Dairy:	78
What Qualifies as Hard Cheese:	80
Do Waiters Need to Wash Their Hands:	80
Dishes Containing Meat or Dairy (תבשיל של בשר או חלב):	81
Utensils of Meat and Dairy:	82
Removing Meat or Dairy Status From a Utensil:	82
טבילת כלים:	83
Metal Knives:	83
Salt Dishes:	83
Salt in Meat/Dairy Utensils:	84
Contact:	84
Bread:	85
Storage:	86
Meliha — Salting	87
מליח כרותח:	87
Levels of Salting:	88
מליח לקדירה:	88
מליח לצלי:	89
Drippings:	89
מליח להצניע:	89
מליח לדרך:	89
How "רותח" is כרותח?	91
Depth of Flavor Transfer:	91
Taarovot — Mixtures	93
ביטול:	93
בליעות & טעם כעיקר:	93
ביטול בששים:	94
מין במינו:	94
קליפה:	95
Exceptions to ביטול בששים:	95
מעמיד:	95
יד סולדת:	96

נותרן טעם לפגם:	96
בן יומו:	97
סתם כלים אינם בני יומן	97
חענ"נ:	97
חתיכה עצמה נעשית נבילה (חענ"נ):	97
Egg Shells:	97
Sharp Foods (דבר חריף):	97
When Sliced With a Knife:	98
Reinvigorating the flavor (מכליא לשבח):	98
Residue:	99
Utensils:	100
Tikun Keilim — Vessel Rectification	102
טבילת כלים:	103
הסרת בליעות:	104
Materials:	104
Earthenware:	104
Glass:	104
Metal:	104
הגעלה:	105
ליבון קל:	105
ליבון גמור:	105
Wood:	105
General Laws:	106
Drippings:	106

Concepts

Bishul — Cooking

Cooking (בישול) is used as both a general term to refer to food preparation and as a specific term to refer to preparing food in heated water (or other fluids, see below).

מראת עין

Whether cooking (a mixture that is intrinsically permitted to be cooked but not eaten, in an instance where an observer seeing this might be misled that eating such a mixture or a similar mixture that is likewise prohibited) can be prohibited because of מראת עין is debated by the ט"ז and ש"ך (יו"ד פז:ד). The ש"ך (ט"ז) holds it can, but the ש"ך (ט"ז) holds that it cannot because any observer can assume that he's cooking the (seemingly) forbidden mixture for medicinal or other non-eating purposes. ^[1]

-
1. The problem with the ש"ך is that his point that seeing someone cook won't lead to the assumption that eating is permitted because there are other valid reasons to cook doesn't help by בשר בחלב because it's prohibited מדאורייתא to *cook* meat and milk together, so even if an observer thinks that the cooking of, say, chicken and milk is for medicinal purposes, we should be worried that he'll conclude that cooking *meat* and milk for medicinal purposes is permitted. R' Yitzchok Dov Lefkowitz told me in the name of R' Heinneman that the ש"ך can be explained as holding like the כסף משנה (see פרק א הלכה ב) who holds that cooking בשר בחלב is only prohibited מדאורייתא if it is being cooked *for the purpose of eating*. Therefore, if someone sees another person cooking chicken with milk, assumes that it's meat and milk, assumes that he's cooking the mixture for medicinal purposes, and therefore concludes that cooking meat and milk for medicinal purposes is permitted, he still won't transgress איסור דאורייתא because he didn't cook the meat and milk for the purpose of eating.

In בשר וחלב:

Cooking:

In the context of בשר בחלב, cooking has two definitions: a general definition of causing flavor to transfer, and a specific definition of cooking in water or other liquids. When the Torah prohibits cooking meat and milk, it means the second definition of cooking in water (שו"ע פז:א) (when heated by a flame, not the sun or hot springs^[1]), and possibly roasting^[2], putting cold meat into hot milk or vice versa^[3], pan-frying^[4], smoking^[5], and cooking in water heated by hot springs or other non-flame sources^[6].

Possible דאורייתא Cooking Methods:

Roasting (צלי):

Roasting differs from cooking^[7] in that the food being roasted is not immersed in liquid, such as on a grill or ungreased griddle. There is some debate whether roasting בשר בחלב is prohibited מדאורייתא or only מדרבנן. The פרי חדש is of the opinion that it is מדאורייתא (see פז ס' פז"ק ב for his proofs.) רעק"א (א) שם brings proofs that the ר"ן argues and holds that roasting is not a דאורייתא method of cooking בב"ח.

Cold into Hot: _____

1. p. 28
2. p. 24
3. p. 25
4. p. 27
5. p. 28
6. p. 28
7. p. ?

תתאה גבר:

When something hot comes into contact with something cold, whether the hot item heats the cold item or the cold item cools the hot item — the consequence being whether the items cook together and thereby transfer flavors to each other — is debated in the Gemara (source). We Pasken that whichever item is lower (although according to many this is actually interpreted to mean the stationary item) extends its state to the upper (moving) item (תתאה גבר), so if the lower (stationary) item is hot, it makes the upper (moving) item hot as well, and flavor transfers throughout both. If, however, the upper (moving) item is hot and the lower (stationary) item is cold, the lower cools the upper and the two items just need קליפה^[1] ^ [The reason הדחה^[2] is not enough, even though they're both cold, is that the cooling is not instant and in the time it takes for the upper item to cool the flavor can transfer כדי קליפה (see ש"ך יו"ד ס' צא ס"ק ז) (see שו"ע יו"ד צא:ד).]

As a consequence of this, when cold milk falls onto hot meat or cold meat falls into hot milk, according to many this would be considered akin to roasting, as the hot base would heat the cold item falling onto it, causing flavors to transfer between them, rendering the mixture בשר בחלב.

Hot into Cold:

If hot meat fell into cold milk, the meat needs קליפה. The law regarding the milk is a matter of dispute. According to the ט"ז (ס' צא ס"ק ז) the milk needs ששים^[3] against the קליפה. According to the ש"ך (שם ס"ק ח) since the קליפה of a liquid mixes in and is not a distinct part, the milk does not need ששים.

1. see Measurements, כדי קליפה, p. ?

2. see Measurements, Surface Level, p. ?

3. see 09539, מתערובות, p. ?

The (The agrees with the ש"ך that the milk does not need ששים, but only when the meat did not enter the milk as part of an uninterrupted stream (עירוי שנפסק הקילוח), so even the קליפה transfer is only at the level of a שני כלי^[1] and is not actually אסור. In such a case, the requirement to remove the קליפה from the meat is only a חומרא due to the optics of eating the portion of the meat that came into contact with milk. Such a concern doesn't apply to the milk because there is no distinct קליפה in fluid[^][In my opinion, the ש"ך would agree to this, although ר' אברהם אליהו לייביקר disagrees with me. This argument hinges on whether the ש"ך in ס"ק ז that says that the ruling of the שו"ע applies to both הקילוח and not is referring just to the קליפה of the meat or also to the milk. It would seem that the (הובא באוצר המפרשים) agrees with him, because he asks how the ש"ך can allow the milk to be consumed without ביטול and therefore disagrees with the ש"ך and explains the רמא as we just have א בשם המג"א].

The (ה) adds that this permit applies to meat and milk, which need to *cook* together to be prohibited מדאורייתא, but not hot איסור that fell into cool היתר, except בהפסד מרובה, because were the בליעות of איסור to transfer, even without cooking, the food would be מדאורייתא אסור.

If cold milk were poured on hot meat, whether the milk is אסור depends on the speed with which it was poured. If it was poured slowly, all of the milk is אסור, because as the stream hits the meat the milk at the tip of the stream pushes aside the milk already in contact with the meat, so that all of the milk directly contacts the meat and is therefore part of the קליפה (see ח"ס פ"ת ס"ק ו who brings from the ח"ס).).

Pan-Frying:

The פרי חדש is of the opinion that it is מדאורייתא (see פז ס"ק ב for his proofs.) The 1. see בישול - Cooking. שני. p. 7. כלי שני. ? brings the מ"י and the מהר"ם שיף who argue and say it is only מדרבנן (שם ס"ק ג).

R' Lefkowitz told me that deep frying is considered cooking, not frying. (Although *in רשי* סנהדרין ד: seems to me to indicate otherwise)

Smoking (מעושן):

There is a debate about what kind of smoking is referred to here. It can either mean hot smoking, in a situation where the heat is indirect, or it can mean cold smoking, i.e. sawdust is burned to create a large amount of smoke in an enclosed space where meat and cheese are in contact, the smoke penetrates the food items and causes flavor to transfer between them. Whether smoking is considered *בישול* in the context of *בשר בחלב* is an undecided debate in the *ירושלמי*, so we consider it a *ספק דאורייתא* (biblical doubt). We therefore treat it like *בישול* due to the rule of *ספק דאורייתא לחומרא* (we rule stringently in a biblical doubt), but because it is only out of doubt the court does not prescribe lashes for such cooking (טו) *מ"מ*, but the *מ"מ* agrees with this (פז:ו). (מגיד משנה הל' מ"א פ"ט ה' ו). The *גר"א* (שם ס"ק דאורייתא) argues that smoking *should* be considered cooking *שבת* ^{^fa19bd} that says smoking is considered cooking regarding *שבת*.

Cooking in Hot Springs (חמי טבריא):

Hot springs (specifically the hot springs of Tiberias) are used as the Talmudic shorthand for heat that is not derived from fire. This category includes geothermal heat and solar heat. It is compared in the *ירושלמי* to *מעושן*, and so shares its laws^[1].

Electricity:

I spoke with R' Yitzchok Dov Lefkowitz, and he told me that whether electric cooking falls under this category depends on the method of heating:

1. p. 28

A directly solar powered kettle (such as the rooftop hot water heaters common in Israel, known locally as a דוד שמש) is considered by some authorities to be indirectly heated by the sun and by some to be directly heated by the sun (with consequences for the laws of שבת), either way such a kettle would fall under the category of non-flame heats.

A standard electric stove is considered by most authorities to be a flame-based heat, because the coil is heated to the point of glowing (like why incandescent bulbs are considered fire for various הלכות).

A microwave and an induction stove are controversial, because R' Moshe Feinstein holds that any normal cooking method is considered בישול, but there are those that argue.

Definite Cooking Methods:

Soaking (כבוש):

Soaking food for 24 hours in liquid causes בליעות^[1] to transfer as if the mixture and the utensil it is contained in were cooked. However, meat and milk that are soaked together are *not* considered to have been cooked together מדאורייתא (see שו"ע צא:ח), as only flame (or heat, depending on the status of חמי טבריא^[2]) derived cooking is included in the biblical prohibition of בשר בחלב. That said, if one were to cook meat that had been soaked in milk or vice versa, that *would* be cooking בשר בחלב מדאורייתא.

That's all as far as דאורייתא is concerned, but if meat and milk were soaked together they would be prohibited מדרבנן to be eaten^[3].

1. see 2, תערוכות, p. 098f.

2. p. ?

3. p. ?

Salting (מליחה):

When salted foods come in contact, בליעות are transferred between them^[1]. Like soaking^[2], this does not constitute cooking מדאורייתא as far as the laws of ב"ח. Therefore, if salted meat and dairy come into contact, they would each be prohibited to be eaten מדרבנן and to be cooked מדאורייתא (since they each contain בליעות of the other).

Normally, salt on a food effects a one-way transfer, meaning that if salted meat came into contact with unsalted dairy, the dairy would be prohibited and the meat would stay permitted (במקום הפסד)^[3]. If the unsalted food was liquid, however, the salted food will absorb some of it as well^[4]. Therefore, if salted meat came into contact with unsalted milk, they would both be אסור.

Meat and Milk Combined Without Cooking:

As discussed above^[5], meat and milk cannot be eaten together, even if not cooked together. This differs from even דרבנן cooking methods in that there is no בליעה transfer at all. This is why, for example, a deli and cheese sandwich would be prohibited to eat, and is the foundation for other important laws, such as Waiting Between Meat and Dairy^[6] and Joint Placement on a Table^[7].

General Contact:

1. see מליחה, How "רותח" is כרותח?, p. ?

2. p. 29

3. see 1, מליחהf3ddb, p. ?

4. see 0, מליחהc7938, p. ?

5. p. ?

6. p. 74

7. p. 72

If cold meat and dairy touched one another: if they were moist, they require הדחה^[1] (see שו"ע צא:א), and if they were dry then nothing need be done because nothing transferred from one to the other (ש"ך שם בשם הב"ח), unless they were salted, in which case they would need הדחה (see רעק"א שם מביא מסע' ה).

Immersion:

If meat[^][assuming it was not salted enough to effect מליחה^[2]] was immersed in milk, there are multiple factors that can cause the meat to absorb milk to the point that הדחה will not be able to permit it, and there are multiple opinions as to which factors or combination thereof cause this absorption. The factors are:

- Roasting - the meat previously being roasted (or baked or cooked, see רמ"א צא:ז) makes it more absorptive according to some.
- Crevices - some meat has deep crevices into which the milk it is immersed in can seep and from which it can't be removed.
- Spicing - if the meat was heavily spiced, it can take on a level of חריפות^[3] from the spice and become more absorptive.

The ש"ך (ס' צא ס"ק כא) lays out the three opinions regarding these factors:

1. see Measurements, Surface Level, p. ?

2. p. 30

3. p. ?

1. The opinion of the ספר התרומה, the שערים, the רא"ש, the טור, the מהרי"ל, and others is that if the meat was roasted then the milk penetrates the surface to the depth of a קליפה^[1]. If the meat had crevices or was heavily spiced, even if it was raw, the milk penetrates deep into the meat and prohibits it entirely.
2. The ראב"ד, the רשב"א, and ר"ן (and possibly רש"י) say that roasting alone has no effect, and cold roasted meat would just need הדחה like any other case of contact^[2], and hot roasted meat would require קליפה, like any case of hot into cold^[3]. If the roasted meat had crevices or was heavily spiced and was hot^[it seems that according to this opinion that if even roasted meat was cold, crevices and spicing would not be an issue.], it is entirely prohibited.
3. The אור זרוע, the שם מהר"ף, the סמ"ק, and many others agree that roasting alone has no effect. However, they say that so long as the meat is roasted, even if it's cold, crevices and spicing would cause it to absorb the milk and become אסור.

1. see Measurements, כדי קליפה, p. ?

2. p. ?

3. p. 26

The (צא"ז) rules like the second opinion, though the (רמ"א) says we should be stringent like the first opinion that even cold roasted needs קליפה, unless doing so will result in a significant financial loss (הפסד מרובה).¹ [The (רמ"א) also says to be lenient (בהפסד מרובה) to consider מליח לצלי^[1] as not רותח. Therefore, according to the (רמ"א), if cold, roasted meat that is מליח לצלי and its קליפה qualifies as מרובה הפסד fell into milk, permitting it would require two leniencies: not considering the meat to be salted to the level of כרותח and not requiring קליפה from cold, roasted meat that fell into milk, and we don't permit combined leniencies (טז שם ס"ק יב).]. The (ש"ך) is hesitant to be lenient even (בהפסד מרובה), because there are so many ראשונים who are stringent.

Even according to the opinions that cold roasted meat absorbs milk and becomes prohibited, we do not assume that the meat exudes any flavor back in to the milk, and so the milk remains permitted (ש"ך שם ס"ק כה).

Dehydrated Meat:

One of the methods used to make cheese is to place a piece of the stomach in the milk, so that the stomach enzymes cause the milk to curdle. This would normally be prohibited as the combination of meat (the stomach) and milk, and if the resulting cheese contains a detectable flavor of meat, the cheese would be prohibited (שו"ע פז:יא).² [The reason the meat is able to be nullified even though it is a מעמיד^[2] is that the stomach is not intrinsically prohibited, it's just its combination with milk that is prohibited.^[3]

1. see מליחה, מליח לצלי, p. ?

2. see תערובות, מעמיד, p. ?

3. see תערובות, 76, p. ?

The פז"י (רמ"א) proposes a solution to this problem wherein one could dehydrate the stomach until it is so dry that it's "like wood and has no moisture of meat." Such flesh can be used to make cheese, as the dry enzymes will still have an effect but there will be no flavor of meat transferred into the milk.

The שם ס"ק לג (ש"ך) argues on the רמ"א and says that one may not intentionally use dehydrated stomach to curdle cheese, although he agrees that if it was used the resulting cheese would be permitted.

He adds that the same dehydration process would also remove the ability to impart flavor from the other intestines.

תתאה גבר:

When something hot comes into contact with something cold, whether the hot item heats the cold item or the cold item cools the hot item — the consequence being whether the items cook together and thereby transfer flavors to each other — is debated in the Gemara (source). We Pasken that whichever item is lower (although according to many this is actually interpreted to mean the stationary item) extends its state to the upper (moving) item (תתאה גבר), so if the lower (stationary) item is hot, it makes the upper (moving) item hot as well, and flavor transfers throughout both. If, however, the upper (moving) item is hot and the lower (stationary) item is cold, the lower cools the upper and the two items just need קליפה^[1] [The reason הדחה^[2] is not enough, even though they're both cold, is that the cooling is not instant and in the time it takes for the upper item to cool the flavor can transfer כדי קליפה (see ש"ך יו"ד ס' צא ס"ק ז (שו"ע יו"ד צא:ד).].

1. see Measurements, כדי קליפה, p. ?

2. see Measurements, Surface Level, p. ?

Cooking After Removal From the Flame (

כלי ראשון/שני/שלישי:
כלי ראשון:

When a pot of food that has been heated to יד סולדת^[1] is removed from the flame, it retains its ability to cook additional food that is placed in it until it cools to below יד סולדת. This is called a כלי ראשון. The (א"ח ס' שיח) disagrees and says that even a כלי ראשון doesn't cook.

כלי שני:

If the food in the pot is moved to different utensil, the food loses most of its ability to cook, *even though it's still* יד סולדת, and, as far as שבת can only cook קלי הבישול, but as far as תערובות can still, according to some, cause בליעות to transfer.

כלי שלישי:

If ^[2]כלי שני food is further moved to a third utensil, it has no ability to cook or transfer בליעות, again *even though it's still* יד סולדת.

1. see תערובות, יד סולדת, p. ?

2. p. 105

Measurements

Depths of Absorption:

Surface Level:

When two items come in contact, they will often leave residue on the surface of the other. If that residue must be removed, such as when kosher meat comes into contact with non-kosher meat and the residue of the non-kosher meat must therefore be removed from the kosher meat so it is not consumed with the kosher meat, the removal can be accomplished by rinsing the point of contact (הדחה), although for some, more oily things (such as rendered fat, see ס' צא ס"ק א), scrubbing the point of contact (שפשוף) is required while rinsing.

גרידה:

When soft, wet surfaces, such as the inside of cucumbers, cannot have substances removed from their surfaces by rinsing, as rinsing such items will not only fail to remove the prohibited substance, it will in fact push it deeper into the surface. The solution in such cases is, therefore, to remove the prohibited substance by means of scraping the surface to remove a very small amount of the surface and the prohibited substance with it (שו"ע צו:ה).

כדי קליפה:

There are certain flavor transfers that only penetrate to a minimal depth within the target, the depth being the thinnest layer of that substance that can be peeled away without falling apart. This is referred to as כדי קליפה (lit. to the point of peeling), and the area of absorption is itself referred to as a כדי קליפה or just the קליפה.

כדי נטילה:

There are certain flavor transfers that penetrate a fingerbreadth into the target. Such cases are referred to as absorbing כדי נטילה (lit. to the point of removal), and the area of absorption is itself referred to as a כדי נטילה or just the נטילה.

Marat Ayin — Appearance

There are many things that should technically be permitted but were prohibited מדרבנן because they give the appearance of wrongdoing. This is called מראת עין (lit. the appearance of the eye).

Applicability to בישול:

מראת עין

Whether cooking (a mixture that is intrinsically permitted to be cooked but not eaten, in an instance where an observer seeing this might be misled that eating such a mixture or a similar mixture that is likewise prohibited) can be prohibited because of מראת עין is debated by the ט"ז and ש"ך (יו"ד פז:ד). The ט"ז (שם ס"ק ה) holds it can, but the ש"ך (שם ס"ק ז) that it cannot because any observer can assume that he's cooking the (seemingly) forbidden mixture for medicinal or other non-eating purposes. ^[1]

1. [footnote 1]

יורה דעה

Basar B'Chalav — Meat and Milk

Note: any citation of a ספר that is divided according to the four חלקים of the טור is referring to יורה דעה unless otherwise specified.

The Prohibition:

The Torah prohibits cooking^[1] meat and milk together, and prohibits the product of such cooking from being eaten or benefitted from (ש"ע פז:א).

The derivation for this law is as follows: the Torah writes (לד:כו, דברים יד:כא) לא תבשל גדי בחלב אמו (Thou shalt not cook a goat in its mother's milk) three times. The פסוק (גמ' חולין קטו:) ברייתא says that the Torah wrote this three times in order to prohibit cooking (1), eating (2), and benefitting from (3) a combination of meat and milk. The ת"ה הקצר ב"ג ש"ד פד: teaches (רשב"א) an important caveat: the Torah only prohibits eating and benefitting from a combination of meat and milk that have been *cooked* together, not just combined. He derives this from the fact that all three פסוקים use the language of "don't cook," even though two of them are referring to eating and benefit, respectively.

It is important to note that this only constitutes an absolute permit to *benefit* from a meat/milk combination that has not been cooked together. *Eating* such a combination, on the other hand, is prohibited מדרבנן. Cooking a meat/milk combination of which one or both components are only considered meat or milk is permitted (in regard to the basic הלכה; see below regarding מראת עין^[3]). (שו"ע פז:ג) As well, benefitting from any meat/milk combination that is only prohibited מדרבנן (whether because of the cooking method or the ingredients) is allowed (רמ"א בהגה פז:א).

1. see בישול - Cooking, 8df37a, p. ?

2. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=115b&format=pdf>

3. see 2017, מראת עין, p. 6

Definitions:

There are three terms that must be defined in order to understand the prohibition of בשר בחלב: cooking^[1], meat^[2], and milk^[3].

Cooking:

In the context of בשר בחלב, cooking has two definitions: a general definition of causing flavor to transfer, and a specific definition of cooking in water or other liquids. When the Torah prohibits cooking meat and milk, it means the second definition of cooking in water (שו"ע פז:א) (when heated by a flame, not the sun or hot springs^[4]), and possibly roasting^[5], putting cold meat into hot milk or vice versa^[6], pan-frying^[7], smoking^[8], and cooking in water heated by hot springs or other non-flame sources^[9].

Possible דאורייתא Cooking Methods:

Roasting (צלי):

-
1. p. 24
 2. p. 34
 3. p. 39
 4. p. 28
 5. p. 24
 6. p. 25
 7. p. 27
 8. p. 28
 9. p. 28

Roasting differs from cooking^[1] in that the food being roasted is not immersed in liquid, such as on a grill or ungreased griddle. There is some debate whether roasting בשר בחלב is prohibited מדאורייתא or only מדרבנן. The פרי חדש is of the opinion that it is מדאורייתא (see פו"ס"ק ב for his proofs.) רעק"א brings proofs that the ר"ן argues and holds that roasting is not a דאורייתא method of cooking בב"ח.

Cold into Hot:

1. p. ?

תתאה גבר:

When something hot comes into contact with something cold, whether the hot item heats the cold item or the cold item cools the hot item — the consequence being whether the items cook together and thereby transfer flavors to each other — is debated in the Gemara (source). We Pasken that whichever item is lower (although according to many this is actually interpreted to mean the stationary item) extends its state to the upper (moving) item (תתאה גבר), so if the lower (stationary) item is hot, it makes the upper (moving) item hot as well, and flavor transfers throughout both. If, however, the upper (moving) item is hot and the lower (stationary) item is cold, the lower cools the upper and the two items just need קליפה^[1] [The reason הדחה^[2] is not enough, even though they're both cold, is that the cooling is not instant and in the time it takes for the upper item to cool the flavor can transfer כדי קליפה (see ש"ך יו"ד ס' צא ס"ק ז) (see שו"ע יו"ד צא:ד)].

As a consequence of this, when cold milk falls onto hot meat or cold meat falls into hot milk, according to many this would be considered akin to roasting, as the hot base would heat the cold item falling onto it, causing flavors to transfer between them, rendering the mixture בשר בחלב.

Hot into Cold:

If hot meat fell into cold milk, the meat needs קליפה. The law regarding the milk is a matter of dispute. According to the ט"ז (ס' צא ס"ק ז), the milk needs ששים^[3] against the קליפה. According to the ש"ך (שם ס"ק ח), since the קליפה of a liquid mixes in and is not a distinct part, the milk does not need ששים.

1. see Measurements, כדי קליפה, p. ?

2. see Measurements, Surface Level, p. ?

3. see 09539, מתערובות, p. ?

The (שם) agrees with the ש"ך that the milk does not need ששים, but only when the meat did not enter the milk as part of an uninterrupted stream (עירוי שנפסק הקילוח), so even the קליפה transfer is only at the level of a שני כלי^[1] and is not actually אסור. In such a case, the requirement to remove the קליפה from the meat is only a חומרא due to the optics of eating the portion of the meat that came into contact with milk. Such a concern doesn't apply to the milk because there is no distinct קליפה in fluid[^][In my opinion, the ש"ך would agree to this, although ר' אברהם אליהו לייביקר disagrees with me. This argument hinges on whether the ש"ך in ס"ק ז' that says that the ruling of the שו"ע applies to both הקילוח and not is referring just to the קליפה of the meat or also to the milk. It would seem that the (הוא) agrees with him, because he asks how the ש"ך can allow the milk to be consumed without ביטול and therefore disagrees with the ש"ך and explains the רמא as we just have א במג"א].

The (ה) adds that this permit applies to meat and milk, which need to *cook* together to be prohibited מדאורייתא, but not hot איסור that fell into cool היתר, except בהפסד מרובה, because were the בליעות of איסור to transfer, even without cooking, the food would be מדאורייתא.

If cold milk were poured on hot meat, whether the milk is אסור depends on the speed with which it was poured. If it was poured slowly, all of the milk is אסור, because as the stream hits the meat the milk at the tip of the stream pushes aside the milk already in contact with the meat, so that all of the milk directly contacts the meat and is therefore part of the קליפה (see ח"ס פ"ת ס"ק ו' who brings from the ח"ס).).

Pan-Frying:

The פרי חדש is of the opinion that it is מדאורייתא (see פו' ס"ק ב' for his proofs.) The 1. see בישול - Cooking. שני. p. 7. כלי שני. ? brings the מ"י and the מהר"ם שיף who argue and say it is only מדרבנן.

R' Lefkowitz told me that deep frying is considered cooking, not frying. (Although in רשי סנהדרין ד: seems to me to indicate otherwise)

Smoking (מעושן):

There is a debate about what kind of smoking is referred to here. It can either mean hot smoking, in a situation where the heat is indirect, or it can mean cold smoking, i.e. sawdust is burned to create a large amount of smoke in an enclosed space where meat and cheese are in contact, the smoke penetrates the food items and causes flavor to transfer between them. Whether smoking is considered בישול in the context of בשר בחלב is an undecided debate in the ירושלמי, so we consider it a ספק דאורייתא (biblical doubt). We therefore treat it like בישול due to the rule of ספק דאורייתא לחומרא (we rule stringently in a biblical doubt), but because it is only out of doubt the court does not prescribe lashes for such cooking (מגיד משנה הל' מ"א פ"ט ה' ו). The טו (מ"מ) agrees with this ש"ע (פז:ו). גר"א (שם ס"ק דאורייתא) argues that smoking *should* be considered cooking, citing a שבת that says smoking is considered cooking regarding שבת.¹

Cooking in Hot Springs (חמי טבריא):

Hot springs (specifically the hot springs of Tiberias) are used as the Talmudic shorthand for heat that is not derived from fire. This category includes geothermal heat and solar heat. It is compared in the ירושלמי to מעושן, and so shares its laws^[1].

Electricity:

I spoke with R' Yitzchok Dov Lefkowitz, and he told me that whether electric cooking falls under this category depends on the method of heating:

1. p. 28

A directly solar powered kettle (such as the rooftop hot water heaters common in Israel, known locally as a דוד שמש) is considered by some authorities to be indirectly heated by the sun and by some to be directly heated by the sun (with consequences for the laws of שבת), either way such a kettle would fall under the category of non-flame heats.

A standard electric stove is considered by most authorities to be a flame-based heat, because the coil is heated to the point of glowing (like why incandescent bulbs are considered fire for various הלכות).

A microwave and an induction stove are controversial, because R' Moshe Feinstein holds that any normal cooking method is considered בישול, but there are those that argue.

Definite Cooking Methods:

Soaking (כבוש):

Soaking food for 24 hours in liquid causes בליעות^[1] to transfer as if the mixture and the utensil it is contained in were cooked. However, meat and milk that are soaked together are *not* considered to have been cooked together מדאורייתא (see שו"ע צא:ח), as only flame (or heat, depending on the status of חמי טבריא^[2]) derived cooking is included in the biblical prohibition of בשר בחלב. That said, if one were to cook meat that had been soaked in milk or vice versa, that *would* be cooking בשר בחלב מדאורייתא.

That's all as far as דאורייתא is concerned, but if meat and milk were soaked together they would be prohibited מדרבנן to be eaten^[3].

1. see 2, תערוכות, p. ?

2. p. ?

3. p. ?

Salting (מליחה):

When salted foods come in contact, בליעות are transferred between them^[1]. Like soaking^[2], this does not constitute cooking מדאורייתא as far as the laws of ב"ח. Therefore, if salted meat and dairy come into contact, they would each be prohibited to be eaten מדרבנן and to be cooked מדאורייתא (since they each contain בליעות of the other).

Normally, salt on a food effects a one-way transfer, meaning that if salted meat came into contact with unsalted dairy, the dairy would be prohibited and the meat would stay permitted (במקום הפסד)^[3]. If the unsalted food was liquid, however, the salted food will absorb some of it as well^[4]. Therefore, if salted meat came into contact with unsalted milk, they would both be אסור.

Meat and Milk Combined Without Cooking:

As discussed above^[5], meat and milk cannot be eaten together, even if not cooked together. This differs from even דרבנן cooking methods in that there is no בליעה transfer at all. This is why, for example, a deli and cheese sandwich would be prohibited to eat, and is the foundation for other important laws, such as Waiting Between Meat and Dairy^[6] and Joint Placement on a Table^[7].

General Contact:

1. see מליחה, How "רותח" is כרותח?, p. ?

2. p. 29

3. see 1, מליחהf3ddb, p. ?

4. see 0, מליחהc7938, p. ?

5. p. ?

6. p. 74

7. p. 72

If cold meat and dairy touched one another: if they were moist, they require הדחה^[1] (see שו"ע צא:א), and if they were dry then nothing need be done because nothing transferred from one to the other (ש"ך שם בשם הב"ח), unless they were salted, in which case they would need הדחה (see רעק"א שם מביא מסע' ה).

Immersion:

If meat[^][assuming it was not salted enough to effect מליחה^[2]] was immersed in milk, there are multiple factors that can cause the meat to absorb milk to the point that הדחה will not be able to permit it, and there are multiple opinions as to which factors or combination thereof cause this absorption. The factors are:

- Roasting - the meat previously being roasted (or baked or cooked, see רמ"א צא:ז) makes it more absorptive according to some.
- Crevices - some meat has deep crevices into which the milk it is immersed in can seep and from which it can't be removed.
- Spicing - if the meat was heavily spiced, it can take on a level of חריפות^[3] from the spice and become more absorptive.

The ש"ך (ס' צא ס"ק כא) lays out the three opinions regarding these factors:

1. see Measurements, Surface Level, p. ?

2. p. 30

3. p. ?

1. The opinion of the ספר התרומה, the שערים, the רא"ש, the טור, the מהרי"ל, and others is that if the meat was roasted then the milk penetrates the surface to the depth of a קליפה^[1]. If the meat had crevices or was heavily spiced, even if it was raw, the milk penetrates deep into the meat and prohibits it entirely.
2. The ראב"ד, the רשב"א, and ר"ן (and possibly רש"י) say that roasting alone has no effect, and cold roasted meat would just need הדחה like any other case of contact^[2], and hot roasted meat would require קליפה, like any case of hot into cold^[3]. If the roasted meat had crevices or was heavily spiced and was hot^[it seems that according to this opinion that if even roasted meat was cold, crevices and spicing would not be an issue.], it is entirely prohibited.
3. The אור זרוע, the שם מהר"ף, the סמ"ק, and many others agree that roasting alone has no effect. However, they say that so long as the meat is roasted, even if it's cold, crevices and spicing would cause it to absorb the milk and become אסור.

1. see Measurements, כדי קליפה, p. ?

2. p. ?

3. p. 26

The (רמ"א) says we should be stringent like the first opinion that even cold roasted needs קליפה, unless doing so will result in a significant financial loss (הפסד מרובה).¹ The (רמ"א) also says to be lenient (בהפסד מרובה) to consider מליח לצלי^[1] as not רותח. Therefore, according to the (רמ"א), if cold, roasted meat that is מליח לצלי and its קליפה qualifies as מרובה הפסד fell into milk, permitting it would require two leniencies: not considering the meat to be salted to the level of כרותח and not requiring קליפה from cold, roasted meat that fell into milk, and we don't permit combined leniencies (טז שם ס"ק יב). The (ש"ך) is hesitant to be lenient even ביהפסד מרובה, because there are so many ראשונים who are stringent.

Even according to the opinions that cold roasted meat absorbs milk and becomes prohibited, we do not assume that the meat exudes any flavor back in to the milk, and so the milk remains permitted (ש"ך שם ס"ק כה).

Dehydrated Meat:

One of the methods used to make cheese is to place a piece of the stomach in the milk, so that the stomach enzymes cause the milk to curdle. This would normally be prohibited as the combination of meat (the stomach) and milk, and if the resulting cheese contains a detectable flavor of meat, the cheese would be prohibited (שו"ע פז:יא).² The reason the meat is able to be nullified even though it is a מעמיד^[2] is that the stomach is not intrinsically prohibited, it's just its combination with milk that is prohibited.^[3]

1. see מליח לצלי, מליחה, p. ?

2. see תערובות, מעמיד, p. ?

3. see תערובות, 76, p. ?

The (פז"י) רמ"א proposes a solution to this problem wherein one could dehydrate the stomach until it is so dry that it's "like wood and has no moisture of meat." Such flesh can be used to make cheese, as the dry enzymes will still have an effect but there will be no flavor of meat transferred into the milk.

The (שם ס"ק לג) ש"ך argues on the רמ"א and says that one may not intentionally use dehydrated stomach to curdle cheese, although he agrees that if it was used the resulting cheese would be permitted.

He adds that the same dehydration process would also remove the ability to impart flavor from the other intestines.

Meat:

Meat is the flesh of a living creature, of which there are five categories as far as the laws of בשר בחלב:

Domesticated Animal (בהמה):

In the context of בשר בחלב, only the meat of a kosher domesticated animal (טהור) is prohibited מדאורייתא. When mentioned unqualified, this is what meat refers to. The meat of a non-kosher בהמה is addressed here^[1].

Blood:

1. p. ?

Blood, although it is prohibited to be consumed^[1], is not included in the laws of בשר בחלב (גמ' חולין קיג:) and is therefore permitted to be cooked with milk and such a mixture is permitted to be benefitted from. (ש"ע פז:ו) There is a debate whether cooked blood is (רמב"ם הל' מ"א ט:ו) or is not (רש"י מנחות כא) included in the prohibition against consuming blood. According to those who permit cooked blood, blood is included in בשר בחלב מדרבנן. Therefore, everyone agrees that it's prohibited to consume blood cooked with milk, either because cooked blood is prohibited or because it is included in בשר בחלב מדרבנן.

Fetus (שליל):

A fetus found in a properly slaughtered domesticated animal is considered full-fledged meat and so has the same laws as a normal domesticated animal^[4].

Amniotic Sac (שליא):

an amniotic sac^[5] is excluded from the laws of ב"ב by the same exclusion as blood^[6].

Quasi-Edible Parts:

The following is a list of animal parts that are only quasi-edible and are therefore not considered to be meat מדאורייתא, although מדרבנן they are still prohibited[^][The ש"ך as interpreted by the פ"ת holds that this prohibition is just for eating, while others argue and say its even for cooking.] (שו"ע פז:ז):

1. <https://www.sefaria.org/Leviticus.3.17?lang=he&with=all&lang2=he>
2. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=113b&format=pdf>
3. <https://www.hebrewbooks.org/shas.aspx?mesechta=30&daf=21&format=pdf>
4. p. 34
5. <https://www.vocabulary.com/dictionary/amniotic%20sac>
6. p. ?

- Skin^[Skin that is exceptionally soft, such as that of the fatty tail, might be considered full-fledged meat (פמ"ג, הובא בפ"ת ס' פז ס"ק טו).]
- Ligaments & Tendons
- Bones
- The soft roots of the horns
- The soft roots of the hooves

Undomesticated Animal (חיה):

The meat of a חיה is consistently grouped with that of birds^[1], and so shares their laws.

Bird (עוף):

Birds are permitted מדרבנן to be eaten with milk, but are treated as meat and are therefore prohibited to be eaten with milk (even of a חיה^[2]). As such, it is permitted to be cooked with milk and for the product of such cooking to be benefitted from. Whether such cooking is actually allowed in practice or is prohibited because of מראת עין is disputed by the (פ"ז ס"ק ה) and the (שם ס"ק ז) ש"ך. The ש"ך says explicitly that it's prohibited, while the פ"ז says that there is no מראת עין^[3] by cooking at all, so obviously he would hold that to cook bird meat with milk is permitted. Unless otherwise specified, when chicken is mentioned in these laws, all kosher birds are included, chicken is just a common example.

Eggs:

1. p. ?

2. p. ?

3. see בישול - Cooking, abd781, p. ?

Eggs that have been laid are permitted to be combined with meat or milk. If they have not yet been laid and were found within a slaughtered chicken, whether they are considered part of the mother, and therefore forbidden to be combined with milk, or like laid eggs that can be combined with milk, depends on what stage of development they have reached.

The Development of Chicken Eggs Within the Mother:

Chickens that are in their egg-laying state (which they are now kept in permanently, by artificial means, so they produce eggs every day) have multiple eggs at various stages of development within them at a given time, each roughly a day ahead of the next. They start as yolks attached by a veiny membrane to the ovary where they grow. When the yolk has grown to full size, the membrane splits, releasing the yolk. The yolk then moves to the next stage where it is surrounded by the white, then a membrane, then the shell.

At What Stage of Development is an Egg Considered an Independent Entity:

The (פז"ה) (שו"ע) says that once the egg has a white, it is considered an independent entity and is permitted to be combined with milk. This would even be true in a scenario where, instead of splitting and releasing the yolk, the membrane detached and remained surrounding the yolk beneath the white. If the egg has not yet developed a white, it is considered meat and cannot be combined with milk. Eating such an egg (so long as the yolk is fully developed) does not, however, require the eater to wait before eating dairy. The (ש"ך) (שם ס"ק יב) says this is true even if the veiny membrane is still on the yolk.

The (שם ס"ק ו) says that the women have a custom to treat even a fully completed egg with a shell to be considered meat if it was found within a slaughtered chicken, and this custom should be followed. However, he says that if the egg was already combined with milk, as long as it has developed the membrane found beneath the shell, it would be permitted. The (ש"ך) (שם ס"ק י) holds that even if the egg has developed the membrane around the white, it still cannot be combined with milk. However, if it has already been combined with milk the mixture is permitted to be eaten, so long as the egg has developed a white and the veiny membrane is not still around the yolk. That said, if prohibiting the mixture would result in a large financial loss, we can rely on those who hold that an egg is permitted to be combined with milk so long as the egg yolk has completely developed, even if it is surrounded by the veiny membrane.

He argues on the (ט"ז), and says that as long as one is not in a place that explicitly follows the women's custom^[1], we are not bound by it.

Fish (דג):

Fish are מעיקר הדין permitted to be eaten with either meat or milk (ש"ע פז:ג), but they are practically prohibited from being eaten with meat due to a סכנה (health risk) (ב"י או"ח סימן קע"ג). The (ש"ך) (פז:ג) adds (ב"י) that there is also a סכנה when they are combined with milk. Because of this, many sects of ספרדים will not eat fish with milk either. The (רמ"א) says (דרכי משה שם ס"ק ד) that this ב"י must be due to a printing error, as the source that the ב"י quotes says fish and *meat*, not milk.

Grasshopper (חגב):

1. p. ?

Technically this category should be insects, but grasshoppers (only certain species^[1]) are the only kosher insects. Because this is the case, there is no halachic term (that I am aware of) that refers to insects and includes grasshoppers as well.

Grasshoppers are like fish^[2] in that they are מעיקר הדין permitted to be eaten with either meat or milk (ש"ע פז:ג), but differ from fish in that there is no סכנה concern associated with their combination with either meat or milk.

Milk:

מדאורייתא:

The only milk that is prohibited under the laws of בשר בחלב מדאורייתא is that of a kosher בהמה. There are milk derivatives, some retain their status of milk and so are included מדאורייתא, and some are considered to have lost their milk status.

Whey:

Whey is a by-product of the cheese-making process. Rennet is introduced to milk, which causes the formation of curds from the fat of the milk. Those curds are then removed and processed further into cheese. The "milk" left behind is called whey.

The (ה)חובא בטור פז:ח) says that whey retains its milk status and is forbidden to be combined with meat מדאורייתא.

דרבנן:

חיה:

בשר בחלב מדרבנן)פ"ת ס' פז ס"ק ז) is included in the laws of חיה.

1. https://ohr.edu/ask_db/ask_main.php/19/Q1/

2. p. 38

In fact, the milk of a חיה is even prohibited to be combined with the meat of a חיה or bird (רמבם הל' מאכלות אסורות ט:ד).

מי חלב:

The step below whey is מי חלב, which is whey that has been cooled off, so the remaining fat floats to the top, and then skimmed, leaving a thin watery substance. This substance is only prohibited to be combined with meat (בטור פז:ח) מדרבנן. (ראש הובא

חלב מתה:

The milk found in a dead animal, whether it was slaughtered properly or it died otherwise (ש"ך פז ס"ק יד), is not included in the prohibition of מדאורייתא בשר בחלב, but it is prohibited מדרבנן to be eaten with meat (ש"ע פז:ו) or cooked with meat (רמ"א בהגה שם). The reason it is not considered milk מדאורייתא is that the גמרא (קיג:) says that the פסוק אמו חלב even though it could have just said חלב to exclude חלב מתה. The גמרא uses this same reasoning to exclude male milk^[2] and the milk of a non-kosher animal^[3].

Rennet:

Rennet is milk found in the belly of a nursing calf. It contains enzymes from the stomach lining which make it a crucial component of the cheesemaking process.

1. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=113b&format=pdf>

2. p. 42

3. p. 44

According to most opinions, when it is found congealed in the stomach it is not considered to be milk, rather it is considered a waste product. It can therefore be combined with meat (שו"ע פז:ט). The (שם) ש"ך adds that although congealed rennet is technically permitted to be combined with meat, one should not intentionally do so because of מראת עין^[1].

Even when the rennet is found in a fluid state, the מחבר says it is not milk, but cites an opinion that it is considered milk and cannot be combined with meat. The (שו"ע שם) רמ"א concurs with this stringent opinion.

Rennet In Cheesemaking:

Rennet that has sat in the stomach of the animal for 24 hours post-slaughter is considered to have been soaked^[2] with it and is therefore prohibited to be used to make cheese, as the בליעות of meat it absorbed from the stomach will be mixed in to the cheese. The same would apply if the stomach was salted before the rennet was removed. (שו"ע פז:י)

The (שם) רמ"א adds that one should not even allow the rennet to remain in the stomach long enough for the stomach to cool down to room temperature, but if one did wait that long the rennet is not prohibited.

If rennet that was left in the stomach for 24 hours was used to make cheese, the status of the cheese depends on the state of the rennet (רמ"א שם):

1. see 6, 2017דמראת עין, p. ?

2. p. 29

1. **The Rennet Was Fluid:** If the rennet was fluid then the cheese is prohibited unless the quantity of milk used to make it was 60 times greater than the quantity of rennet used.^[1]
2. **The Rennet Was Found Congealed:** If the rennet was found congealed in the stomach it is considered waste and therefore even the meat flavor it absorbed is considered to be waste. Because of this, the cheese made with it is not prohibited.
3. **The Rennet Was Found As A Fluid But Later Congealed:** The primarily accepted opinion is to treat such rennet as fluid and therefore prohibit cheese made therewith. There are, however, opinions who treat such rennet as congealed, therefore in a case where prohibiting the cheese will constitute a significant loss we are lenient and permit the cheese.

Male Milk:

The milk of a male animal that has a genetic abnormality which causes the animal to have udders (see רש"י חולין קיג: ד"ה ולא בחלב זכר^[2]) is not included in (ש"ע פז:ו) [בשר בחלב מדאורייתא^[3]]. However, מדרבנן it is included in the prohibition.

מראת עין:

There are various "milks" that are prohibited due to מראת עין, however the specifics of that prohibition is different for each:

Almond Milk:

1. see תערובות, ביטול בששים, p. ?

2. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=113b&format=pdf>

3. p. ?

Almond milk, while not truly milk at all, has enough of an appearance of milk that the (פז:ג) requires meat combined with almond milk to be served with almonds (or some other indicator) nearby to show that the "milk" is not truly milk. The (שם ס"ק ד) extends this to include chicken with almond milk, but he holds that if you don't have almonds, you can still eat the chicken (ב"ה שם ס"ק ו) or the meat (פ"ת שם ס"ק י). The (ש"ך) (שם ס"ק ו) however, disagrees with this ט"ז, based on a מהרש"ל, and holds that without an indicator it would be prohibited to eat even a mixture of *chicken* and almond milk. The (שם פ"ת) adds that chicken with almond milk would only require an indicator if it was served at a large meal, where there would be a real concern of מראת עין, but in a private setting an indicator would not be required. R' Yitzchok Dov Lefkowitz told me in the name of R' Heinneman that since non-dairy "milks" have become so widespread, there is no longer an issue of מראת עין, because an observer that sees a mixture of meat and a milk alternative will assume that it's not real milk. Because of this, the Star-K allows restaurants under its supervision to serve non-dairy milk in an unmarked metal container, as opposed to their old policy of requiring the milk to be brought out in its original container that clearly says "almond milk."

Mother's Milk:

Human breast milk, while it is both a true milk and permitted to be consumed even by adults (once it has been separated from its source), is not included in the laws of בשר בחלב. However it is prohibited (at least with meat, see the discussion of almond milk regarding combinations with chicken^[1]) under the laws of עין מראת and, unlike almond milk, there is no solution available of placing an indicator next to the dish. However such milk is nullified in a simple majority so long as it cannot be seen in the dish. (שו"ע פז:ד)

1. p. ?

Human Male Milk:

In the event of an abnormality which causes a man to produce milk, it would share the laws of mother's milk. (ש"ך as explained by the רמ"א פ"ז:ו)

Milk of a Non-Kosher Animal:

The milk of a non-kosher animal is not prohibited under the laws of בשר בחלב to be combined with meat.^[1] Of course such a combination (or the reverse, non-kosher meat with kosher milk) is forbidden to be eaten, but that is just because the milk was already prohibited to be eaten as the milk of a non-kosher animal. It is, however, permitted to cook and benefit from such a mixture (שו"ע פ"ז:ג). The ש"ך (שם סעיף ד') says that it is forbidden to cook such a mixture because of עין ס"ק ה' (ה), although he does not extend this to chicken/non-kosher milk. The רמ"א (שם ט"ז) considers the רמ"א's reasoning to be flawed but is hesitant to permit that which the רמ"א prohibits. The ש"ך, on the other hand, holds that cooking cannot be prohibited as עין מראת עין at all^[2], so he disagrees with the רמ"א and permits such cooking.

1. p. ?

2. see בישול - Cooking, abd781, p. ?

Laws:

חתיכה עצמה נעשית נבילה (חענ"נ)

Sharp Foods (דבר חריף) and Knives:

Sharp Foods (דבר חריף):

Sharp foods, either intrinsically sharp foods like radish and beets or mixtures consisting of at least a majority sharp foods (רמ"א יו"ד צה:ב), have special laws when it comes to the transfer of בליעות:

What's Considered חריף:

Radishes and beets are considered חריף (certain varieties) (שו"ע צו:א), but their leaves are not¹[The רמ"א in ת"ח brings an או"ה that the leaves of radishes and beets are considered sharp, just to a lesser extent than the radish itself, therefore they absorb flavor from the knife כדי קליפה¹]. (ש"ך ס' צו ס"ק יא). Garlic, onions²[Small, sweet onions (presumably either shallots or Vidalia onions) are not fully חריף, and therefore will not be מכליא לשבח in a scenario that would lead to loss (פ"ת ס' צו ס"ק ג)]. [Onions that are cooked (presumably sauteed as well) are not considered חריף anymore, as the cooking sweetens them (פ"ת ס' צו ס"ק ד)], leek, and similarly sharp foods, along with sour (acidic) fruits³[The רמ"א in ת"ח rules that sour apples are not considered חריף. Therefore, in a situation where ruling apples to be חריף will result in a loss, we can rely on the ת"ח that an apple is not חריף, in conjunction with the opinion that only חילתית is a ס' צו ס"ק ט]. (דבר חריף) and salted fish, are all considered רמ"א (שם). The חריף (שו"ע שם סע' ב) adds horseradish and ginger to the list⁴[Presumably the שו"ע agrees and just included it in "similarly sharp foods."].

1. see Measurements, כדי קליפה, p. ?

The (ש"ך שם ס"ק יח) rules that spices (and salt, שם סע' ג) are considered חריף, and will therefore absorb flavor from the grinder when ground, although only if the grinder is בן יומו. The (רמ"א שם) brings an opinion (like which we rule, ש"ך שם) that spices are חריף enough to absorb flavor, and become אסור/meat thereby, even if the grinder was not בן יומו.

When Sliced With a Knife:

Sharp Foods (דבר חריף):

Sharp foods, either intrinsically sharp foods like radish and beets or mixtures consisting of at least a majority sharp foods (רמ"א יו"ד צה:ב), have special laws when it comes to the transfer of בליעות:

When Sliced With a Knife:

Slicing sharp foods, such as radishes, with a knife (even if it's cold and clean) causes the knife to expel the בליעות that it contains כדי נטילה^[1] from the place of contact with the knife into the radish (ש"ע יו"ד צו:א). This expulsion of בליעות, which normally requires heat, is effected by the combination of the pressure exerted by the knife against the food and the sharp flavor of the food (שם ס"ק ב) (ש"ך).

Consequently, if the knife contained was used for hot meat within 24 hours, or even if it was used more than 24 hours earlier but it still had meat residue on the surface, the radish would not be able to be eaten with dairy until it had a נטילה removed, or if he tasted the radish and was unable to detect flavor of meat, at which point it can just be rinsed^[2] (ש"ע שם).

1. see Measurements, כדי נטילה, p. ?

2. see Measurements, Surface Level, p. ?

The שו"ע brings an opinion that even if the knife was clean and had only been used for hot meat more than 24 hours earlier the radish will still need to have a נטילה removed before it can be combined with dairy. This is because the sharpness of the radish reinvigorates the flavor (מכליא לשבח^[1]) like חלתית does (ו) (ש"ך שם ס"ק).

The שו"ע continues that if one didn't remove the נטילה and cooked the radish with dairy, if the total contents of the pot is greater than 60 times the נטילה or the blade of the knife, whichever is smaller[^][If the blade is smaller than the נטילה, then we need 60 times the blade, because since the blade is the source of the meat flavor in the נטילה, the נטילה can't contain more meat flavor than could have been contained in the knife, i.e. up to the volume of the blade. If the נטילה is smaller than the blade, then we need 60 times the נטילה, as the נטילה can't hold a volume of בליעות that is greater than its own volume.], the mixture is permitted. If not, the mixture will be rendered בב"ח.

The שו"ע ends off that the above would all be true for a knife that contains בליעות of איסור as well. The שו"ע (ט"ז) שם ס"ק ו) says that we, who argue on the שו"ע and hold that חנ"נ applies to all prohibitions, not just בב"ח^[2], rule that even if the blade was smaller than the נטילה, if the radish gets mixed with permitted food, one would need 60 times the נטילה, because when the radish was sliced the בליעות of איסור prohibited the whole נטילה due to חנ"נ.

The שם (רמ"א) adds that if the radish was diced (so each piece is only the size of a נטילה), then one would need 60 times the whole radish. The שם (ש"ך) elaborates:

1. see תערוכות, Reininvigorating the flavor (מכליא לשבח), p. ?

2. p. ?

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- If the knife contained flavor of meat/milk, the above would only be true if the blade had a greater volume than the radish, so there would be enough בליעות in the knife to permeate all of the radish. Otherwise, if one cooked the whole radish with dairy, one would only need 60 times the blade. If we no longer have access to the knife and we don't know its volume, we assume that it had a greater volume than the radish. If one combined a piece of the radish with dairy, we have to assume that the piece was fully permeated with meat flavor^[Even if there was not enough flavor in the knife to permeate the whole radish, since we don't know how the flavor distributed, when dealing with individual pieces we must account for the possibility that the distribution of flavor disproportionately affected the individual pieces we're dealing with.] and the dairy therefore needs to be 60 times larger than the piece of radish.
 - If the knife contained flavor of איסור, then we need 60 times the whole radish, even if the knife was smaller than the radish. This is because we have to assume that the איסור distributed just enough flavor to each piece of the radish to render that piece prohibited (meaning each piece would absorb more than 1/60 of its volume of איסור flavor), and then חז"ל would say that we need to treat each piece as fully prohibited.

The חריף (רמ"א) brings an opinion that even if one only made a single slice in a דבר (e.g. he cut a radish in half), the sharpness of the food draws the flavor throughout the דבר חריף, rendering the whole item אסור (or meat/dairy, depending on what was absorbed in the knife). The רמ"א rules like this opinion, לכתחילה, but says that בדיעבד [As to what is considered בדיעבד here, the ש"ך (ס' צ"ו ס"ק י) says that if it was sliced its still considered לכתחילה, only if it was then cooked with meat/dairy is it considered בדיעבד. The מ"י (הובא ברעק"א שם) says that in the case of a knife of איסור, if the radish was sliced it's בדיעבד, except in a case where it was sliced by a non-jew and there are other radishes available to buy, such a radish would be considered לכתחילה and therefore fully אסור.] the radish is only אסור כדי נטילה.

If the Status of the Knife is Uncertain:

If we are not sure if the knife used to slice the דבר חריף contained איסור or not, the דבר חריף is permitted^[1] [This permit relies on the fact that סתם כלים אינם בני יומן (even though we have a rule that a knife is presumed to be unclean, that's a small amount of איסור and so we don't need to take it into account, (מ"י הובא ברעק"א שם), in conjunction with the fact that many opinions rule that חילתית is the only חריף דבר that is מכליא לשבח. If we knew that it was בן יומו, and we were only unsure if it was used for איסור or if the אסור knife was the knife used, this case would be a ספק דאורייתא and we would rule stringently (ש"ך ס' צו ס"ק יב)]. Therefore we may buy דברים חריפים, such as radishes or ginger, that were harvested by non-jews, because there's a good chance that the radishes were harvested using a utensil that is dedicated for harvesting, rather than a general-purpose knife that presumably contains בליעות of איסור. If we know the radish was cut with a knife, then if there are no other radishes available we may buy from the non-jew and just remove a נטילה (because this is considered בדיעבד [If there were other radishes available to buy, this would be considered לכתחילה and one would not be allowed to buy the radishes from the non-jew (ש"ך ס' צו ס"ק טו)], see above^[2]) (רמ"א צו:ב).

1. see תערוכות, סתם כלים אינם בני יומן p. ?

2. p. ?

Similarly, one may buy lemon juice or salted fish from a non-jew that the non-jew "brought in barrels" (i.e. made in bulk) (שו"ע שם סע' ד). This because even if the non-jew used a knife of איסור, the איסור will become בטל in the large volume of lemons/fish^[only if the lemons were halved (sliced once). If they were quartered (sliced twice, with the second slice bisecting the first) the איסור will never become בטל, because the knife will emit איסור into the lemon with the first slice and that section of the lemon will become חנ"נ. When the second slice passes through that area, the new חנ"נ איסור will be absorbed by the knife and thereby imparted to the next lemon, where the cycle will repeat itself.]. Therefore, we may buy chopped (fermented^[ב"ק יב"ט"ז שם ס"ק יב]) cabbage (presumably sauerkraut) from non-jews, although there are places where the custom is not to buy this cabbage from a non-jew and one should not vary from the custom of his area. Any foods that are not (so^[ג"כ יג"ט"ז שם ס"ק יג]) acidic, like apples may be bought when produced in bulk (רמ"א שם).

Slicing Non-חריף Foods:

If one slices a non-חריף food, like a cucumber, with a meat knife, there are no בליעות transferred, and so the cucumber may be eaten with dairy. If the knife was dirty, the food needs to be cleaned of residue before it can be eaten with dairy, so something moist and soft like a cucumber needs to have the area of the slice scraped away^[Fish that is recognizably kosher (such as salmon) but was sliced with a knife that was used for non-kosher fish may be scrubbed at the point of contact with the knife and it is permitted (פ"ת ס' צו ס"ק ה).], while something hard like a turnip^[לפת in the שו"ע, which the ריבי"ן ש"ך שם ס"ק כב translates as rüben in german, which translates to turnip.] only needs הדחה^[1] (שו"ע צו:ה).

Turnip:

1. see Measurements, Surface Level, p. ?

Turnip has a special property, namely that if one slices a turnip with a meat knife and then slices a דבר חריף like a radish, the radish will not absorb meat flavor because the turnip residue on the knife neutralizes the חריפות of the radish (שם (שו"ע), but only on the first slice immediately following the slicing of the turnip (רמ"א שם). That said, one cannot rely on this לכחילה, meaning if one has a meat knife and wants to cut חריף dairy, he may not slice a turnip and then slice the dairy, but if he does, the dairy will be permitted (ש"ך שם ס"ק כב).

To Which Prohibitions Does חענ"נ Apply:

According to basic Torah law, when איסור is combined with היתר (such as נבילה falling into cholent), the resulting mixture is prohibited, so long as the איסור is not בטל^[1]. It is important to note that the היתר itself does not become אסור, it just becomes inextricably linked with the איסור (flavor of the נבילה distributes throughout the cholent) so that it practically cannot be consumed without also consuming the איסור.

However, when meat and milk are cooked together, they each become prohibited intrinsically, not just due to the meat/milk present in the milk/meat. This is because both the meat and the milk are permitted substances, so if the mixture is declared prohibited, the components must necessarily become prohibited themselves. This concept is known as חתוכה עצמה נעשית נבילה (חענ"נ) (lit. the piece itself becomes נבילה, i.e. intrinsically prohibited) (שו"ע צב:ג-ד).

There is a major dispute amongst the ראשונים whether there is a הרבנן extension of חענ"נ from בב"ח to all other prohibitions. The (שם ב"י) rules that there is no such extension and the רמ"א (see סע' ד) (and therefore this work, as it is written from an Ashkenazic perspective) rules that there is.

The Ramifications of חענ"נ:

The primary consequence of חֶעֱנִי"נ occurs when the prohibited mixture (a cholent containing נבילה meat) itself is mixed into a larger permitted dish (a chicken soup, for example). Without חֶעֱנִי"נ, the larger dish need only contain 60 times the original איסור, and in fact the permitted component can be counted towards that 60 (the soup and the cholent together need to be 60 times the נבילה). With חֶעֱנִי"נ, the permitted component of the original mixture becomes intrinsically prohibited, requiring the second permitted dish to be 60 times the entire first mixture, not just the original איסור (the soup must be 60 times large than the entire cholent, not just the נבילה (שו"ע צב:ד)).

מין במינו and חֶעֱנִי"נ:

According to the שו"ע^[1], if meat fell into boiling milk, it can be given to a non-jew to taste. This is only if the meat was taken out before it had a chance to expel the milk it absorbed, which is when the milk stops boiling. If it was not removed before then, then חֶעֱנִי"נ tells us that the milk that was expelled by the meat is prohibited, but it is מין במינו in the rest of the milk, so it cannot be tasted and the pot therefore needs ששים against the milk (which we assume has the same volume as the meat). The רמ"א says that we don't rely on a non-Jew's tasting and so we need ששים regardless. [As to why we don't need ששים against twice the volume of the meat (the meat itself and the milk that became בשר בחלב), it was explained to me by ר' אברהם אליהו לייבסקר that in order for the milk to enter the meat, the meat must necessarily expel some of its own substance, which would be בטל as it emerges, therefore the final total of prohibited substance that needs to be nullified when the boiling ceases cannot exceed the original volume of the meat.].

חֶעֱנִי"נ Localized

I. see תולדותיו, uc9904, p. ?

If a drop of milk falls into a pot of meat and we know which piece of meat it fell on — if the piece of meat is not large enough to nullify^[1] the milk without the assistance of the rest of the food in the pot, the piece will become *בב"ח* and *חענ"נ* will require the rest of the pot to contain *ששים* against the piece of meat upon which the milk landed. This is only true if the contents of the pot were not stirred[^][because the stirring distributes the milk throughout the pot before it has a chance to be absorbed into the meat it landed on] or covered[^][because when a boiling pot is covered, the intense steam makes even what's above the surface of the liquid as if it's submerged] "from beginning to end, rather they were only stirred/covered at the end," but if they were stirred/covered from beginning to end, the whole pot would combine to nullify the drop of milk (*שו"ע צב:ב*).

The *ט"ז* and *ש"ך* have many disagreements about the above paragraph, most of which stem from the following disagreement between *רש"י* and the *ר"י*:

- According to *ר"י*, the above is referring to where the piece of meat upon which the milk landed was not at all submerged in the liquid in the pot, rather it was resting atop another piece of meat. If it had been even partially submerged, the milk would have passed through the meat into the pot and distributed throughout.
- According to *רש"י*, the above can even be when the meat was partially submerged; since the upper portion of the meat is unsubmerged, the milk doesn't flow[^][shloop] through the meat into the rest of the pot, rather it prohibits the meat upon which it landed, which then exudes its now *אסור* flavor into the rest of the pot.

1. see תערוכות, בליעות & טעם כעיקר p. ?

The רש"י (רש"י) says that the שם ט"ז (שם ט"ז) is written according to רש"י, because we must be discussing a case where the meat was partially submerged. Otherwise, we wouldn't need to have enough in the pot to nullify the meat, we'd just take it out.

The נקודות הכסף (שם) argues with the ט"ז and says that the שם ט"ז is going like the ר"י. As to why we need to nullify the meat, we're simply discussing a case where the pot was mixed after the milk fell on the meat.

As far as stirring/covering, both the ט"ז and ש"ך agree that when the שם ט"ז says "from beginning to end" he means that it was stirred at "the beginning" — when the milk first fell in — but not necessarily that he must continuously mix it until "the end" — when the pot cools off and flavor stops transferring. The שם ט"ז refers to it as "beginning to end" only to contrast with when the pot is stirred at the end only, which is not enough to avoid דמע"נ. This is also the opinion of the רמ"א there.

The רמ"א (שם) says that if he didn't stir the pot at all until it cooled off, if the pot has 60 times the milk then only the piece upon which the milk landed is prohibited.

Both the ט"ז and ש"ך ask that if the flavor isn't leaving the piece — as evidenced by the fact that the pot need only have 60 times the *milk* — why do we need to nullify the milk at all? It remains within the piece it landed on!

The ש"ך answers that if the piece was fully unsubmerged, the pot would not need 60 times the milk. The רמ"א is instead referring to a case where the meat was partially submerged, and he is unsure whether we rule like the ר"י or רש"י, i.e. whether the milk stays within the piece or if it flows through into the pot, so as long as there's 60 times the milk in the pot, the pot is permitted either way. רעק"א answers that the רמ"א holds in general that we always need to suspect that the meat was fatty and so in this case it could be the meat was fatty and the milk was transported through the meat and into the pot, so therefore the pot needs ששים against the milk.

If, however, it was *not* known which piece of meat the milk fell on, the שו"ע rules that one can stir the pot and then the milk is nullified so long as the mixture contains 60 times the milk. The רמ"א argues that mixing it after it fell in doesn't help, it needs to be immediate (and so this presumably would be subject to the laws of being מבטל יבש ביבש, although the ט"ז brings an opinion that we don't say חענ"נ when we don't know which piece it landed on).

Secondary Flavor Transfer (נותן טעם בר נותן טעם):

We know that food cooked in a utensil transfers its flavor into the utensil^[1], that food subsequently cooked in that utensil receives that flavor from the utensil^[2], and that the flavor being present is as if the food it came from is present^[3].

Therefore, if one cooks meat in a pot and then cooks dairy in that pot (within 24 hours^[4], see רמ"א צה:ב), even if the pot is clean, the dairy will receive flavor of meat and be prohibited as בב"ח unless the dairy has a volume that is 60 times that of the meat flavor that was contained within the walls of the pot^[5]

^[Assuming one knows what volume of meat was absorbed in the walls, such as if one cooked a known quantity of meat in the pot, then the walls can't contain more than that quantity of meat. If one doesn't know the volume of the meat flavor in the walls, he must calculate the necessary 60 against the full volume of the walls of the pot (which, in this particular case, means ביטול is virtually impossible, as the dairy contents of the pot can only be 60 times the volume of the walls in a pot that is exceptionally wide with vanishingly thin walls, see ס' צג

[.].(ט"ז ס' צה ס"ק ו) (ב"י).

1. see 2, תערוכות, 098f, p. ?

2. see 4, תערוכות, 4a72, p. ?

3. see 10325, תערוכות, p. ?

4. see תערוכות, בן יומו, p. ?

5. see 09539, תערוכות, p. ?

Such a scenario would be called a prohibited secondary flavor transfer (דאיסורא) (נותן טעם בר נותן טעם): a primary flavor transfer from the meat to the pot and a secondary transfer of that same flavor from the pot to the dairy, which results in a prohibition, i.e. בב"ח. There is a special law that in the event of a *permitted* secondary flavor transfer (נותן טעם בר נותן טעם דהתירה), like the above case but with fish instead of dairy (Meat>Pot>Fish), the meat flavor is considered to have been weakened to the point of exclusion from טעם כעיקר^[1] (as far as לא תבשל^[2] is concerned, see below^[3]) and the fish can therefore be eaten with dairy[^][And the fish itself can be eaten, according to the רמ"א^[4] even though it would be prohibited due to סכנה had it absorbed the flavor directly from the meat^[5].]^[At least as far as the laws of בב"ח. According to the מחבר himself, any combination of fish and dairy is practically prohibited due to סכנה^[6].], even though the fish would not be permitted to eat with dairy had it absorbed the same quantity of flavor directly from the meat, such as if the pot wasn't cleaned well^[Meaning if one *knows* that the pot wasn't cleaned well. If he doesn't know either way, he can assume that it was clean (ש"ך ס' צה ס"ק א) and there was meat residue in the pot amounting to more than 1/60 of the fish (שו"ע צה:א).

1. see 10325, תערוכות, p. ?

2. p. ?

3. p. ?

4. p. ?

5. p. ?

6. p. ?

The reason for this is that we consider each time the flavor transfers to weaken it somewhat, and two such weakenings are enough to exclude the flavor from כעיקר טעם^[1] as far as לא תבשל^[2] is concerned, because flavors that have transferred twice are not considered a normal part of cooking. The (מש"ז צה:א) writes that this is only true if the order is specifically into a utensil and then into food (Meat>Pot>Fish>Dairy), but the other way around (Meat>Fish>Pot>Dairy) would not create נב"נ because we say טעם כעיקר in food until it gets weakened by transferring into a כלי. If the second transfer is creating בב"ח (Meat>Pot>Dairy), then at the point when it mixes with the dairy it has not yet fully transferred twice, and is therefore still considered a normal part of cooking and can create בב"ח. This whole concept only applies to laws of בב"ח, and only before the meat and dairy have combined to become בב"ח. When we are dealing with other prohibitions, such as נבילה ^ [an animal that died without being slaughtered properly, which is prohibited to be eaten מדאורייתא] or בב"ח that has already become prohibited, there is no concept of weakening, because prohibitions like נבילה don't need to meet the standard of being a normal part of cooking, they just need to be present and detectable (שך ס' צד ס"ק כב).

Which Methods of Transfer Effect נב"נ:

As we saw above, the (שו"ע צה:א) holds that נב"נ applies no matter how the flavor transfers are effected.

1. see 10325, תערוכות, p. ?

2. p. ?

The (שם סע' ב) cites an opinion that when both of the flavor transfers are accomplished by cooking^[1] or roasting^[2], such as in the above case of Meat>Pot>Fish, we would not consider it נב"נ, because such methods of transfer have enough "power", as it were, to transfer flavor without diminishing it to the point of נב"נ.

According to these opinions, we would only say נב"נ when the second transfer was the lesser level of placing יד סולדת fish on a cold plate that had previously absorbed בליעות from meat (Meat>Plate>Fish), or vice versa if the fish was cold and the plate was hot (ט"ז שם ס"ק ג).

1. p. ?

2. p. ?

The (לכתחילה) שם (רמ"א) rules that we should be stringent like this opinion initially (לכתחילה), and therefore, in a case of Meat>Pot>Fish for example, not combine the fish with dairy^[Even according to the שו"ע, one would not be permitted to cook the fish in the meat pot with the intention of eating it with dairy, they only argue once the transfer has happened whether you can eat the fish with dairy (ג' (ש"ך ס' צה ס"ק).], but that if one already combined the fish with dairy (fried it in butter, for example) one is allowed to eat it (בדיעבד)^[Only if the one of the transfers was cooking (in water), because then the water acts as a kind of third flavor transfer (ex. if the meat was cooked in the pot: Meat>Water>Pot>Fish). If both transfers were effected by roasting, then we are stringent like these opinions even בדיעבד. Therefore, in a case where the meat was roasted and the fish was cooked (Meat>Pot>Water>Fish), the water of the cooking would itself be prohibited to combine with milk, because it will have absorbed the meat flavor directly from the pot, like roasting (פ"ת ס' צה ס"ק ב).]. In a case of Meat>Plate>Fish, one could even combine the fish with dairy לכתחילה, because even the stringent opinion cited by the רמ"א agrees that such a case is נב"נ^[Although one cannot cause such a transfer of putting hot fish on a meat plate if he's planning on eating the fish with dairy (ט"ז ס' צה ס"ק ד). This stringency would only apply to eating the fish with dairy, but even the stringent opinion would agree that to place נב"נ fish (Meat>Pot>Fish) on a dairy plate would be permitted.

Sharp Foods (דבר חריף):

Sharp foods are an exception to נב"נ, as their sharpness allows them to more powerfully draw the בליעת from their source (the pot, or the grinder, in a case of spices that were ground with a grinder that contains meat בליעות), effecting the transfer without weakening the בליעות even if the pot or grinder are not בני יומן^[1]. Therefore, if the sharp food or spices fall into dairy, the mixture must contain 60 times the sharp food or spices^[We only need 60 times the spices, because although spices are not normally בטל in 60 because their flavor is still detectable at far greater ratios, these spices are not intrinsically prohibited when combined with dairy, only the meat flavor they contain, which is בטל in 60 (ש"ך ס' צה ס"ק ח) in order to be permitted (רמ"א צה:ב).

Washing Dishes:

1. see תערוכות, Reinigorating the (מכליא לשבח) איסור, p. ?

If one washes a meat utensil and a dairy utensil (both of which are בני יומן^[1] ^[Meaning he *knows* that they're בני יומן. If he's unsure, then we can assume they were not בני יומן and permit them, even according to the רמ"א^[2] (ס' צה ד"ה וכמו) in a basin with יד סולדת^[3] water, as long as neither of the utensils have any food residue of meat or dairy on them, they are both permitted, because any flavor transfer between them is נב"נ (Meat/Dairy>Utensil>Water>Utensil) . However, if there was residue of meat or dairy on one of the utensils, both utensils are prohibited (unless the water is 60 times the residue), because the residue of, say, meat will prohibit the dairy utensil (Meat>Water>Dairy Utensil), which will then in turn emit בליעות of בב"ח^[4] [Which, being איסור, are not subject to סע' ג) (שו"ע שם).

1. see תערובות, בן יומו, p. ?

2. p. ?

3. see תערובות, יד סולדת, p. ?

4. p. ?

The רמ"א (שם) argues, and rules that both utensils are prohibited, even in a case where both of them are clean (assuming they are both בני יומן). The reason is either that we are worried that the utensils will make direct contact and flavor will transfer from one to the other and be נב"נ דאיסורא^[1] (Meat/Dairy>Utensil>Utensil(containing Dairy/Meat)), or that we are worried that the meat and dairy בליעות will combine immediately in the water, rendering the water אסור, which in turn makes the utensils ח"ק שם ט"ז. The רמ"א adds that we only prohibit the utensils if they were washed simultaneously and in a כלי ראשון^[2], but if they were washed sequentially (meaning they were never in the water at the same time), or the hot water in the basin was a כלי שני^[3], they would be permitted, because:

- If they weren't in the water at the same time, the two utensils can never make direct contact, and the meat/milk flavor emitted from the first utensil will have come to rest in the water as נב"נ by the time the second utensil is placed in the water; therefore even if the meat/milk flavor then mingles with the milk/meat flavor emitted by the second utensil, the flavors won't combine to create a prohibition.
- If the water was a כלי שני, then, being that we rule like the opinion that a כלי שני does not have the ability to transfer בליעות, the water can't draw the meat and milk flavor out of, or impart flavor into, the utensils in order to create איסור. This would even be true if the utensils were dirty, the כלי שני has no ability to cause the בליעות to move, and so this is no worse than combining cold meat and milk together (ט"ז שם ס"ק יב).

1. p. ?

2. see בישול - Cooking, כלי ראשון, p. ?

3. see בישול - Cooking, כלי שני, p. ?

Even according to the שו"ע, one should not wash meat and milk dishes together לכתחילה, even if they are not בני יומן, because both נב"נ^[1] (see footnote) and נט"ל^[2] are only permitted בדיעבד.

If a dairy utensil was found amongst meat utensils (or vice versa), we are not concerned that they may have been washed together in a way that would prohibit the dairy utensil, and it is therefore permitted (רמ"א שם).

Ashes (Dish Soap):

In a case of washing a meat utensil and dairy utensil together, if one added ashes to the water before adding the utensils, the utensils do not become prohibited, because the ashes render the water, and any בליעות they subsequently absorb from the utensils, נט"ל^[3] (the same would apply nowadays to dish soap, R' Lefkowitz שם ס"ק ד).

The שו"ע (ט"ז) and שם ס"ק כא) both argue on this ruling:

1. p. ?

2. see תערובות, נותן טעם לפגם, p. ?

3. see תערובות, נותן טעם לפגם, p. ?

- the שו"ע shows that the רמ"א must argue from his ruling^[1] prohibiting using a shampoo pot for cooking food, even though that pot is used to cook shampoo which contains ashes and should therefore be permitted because of נט"ל. He additionally found a מהרש"ל בשם מהרי"ל that, in a scenario where a minor reported that soap made from ashes and prohibited fats (חֵלֶב) fell into a pot of food, ruled that the pot is permitted because we do not believe the minor's report. The implication of the ruling is that were we to believe the report, the food would be prohibited, even though the soap contained ash and therefore should have been considered נט"ל according to the שו"ע.
- The שו"ע asks from a תוס' in ד"ה בשקדום (ק. חולין) that is discussing the following: when the Jews defeated the Midianites and plundered their belongings, they were commanded to purify the Midianite's utensils through הגעלה^[2]. The issue תוס' is discussing is how they could have done הגעלה to the largest utensil (that therefore could not be placed inside a larger utensil), the water of the הגעלה would not have been 60 times the flavor that would leave the walls (because it is nearly impossible to have a utensil whose contents are 60 times greater than the volume of its walls), and therefore the flavor would be able to reenter the walls and prohibit the anew. The question of the שו"ע is: if this method of the שו"ע works, why didn't תוס' simply explain that they added ashes to the הגעלה water, rendering any flavor that emerges נט"ל and unable to re-prohibit the pot?

1. p. ?

2. see תיקון כלים, הגעלה, p. ?

The (שם ס"ק ו) answers for the שו"ע that ash does not have the ability to render prohibited flavor נט"ל and therefore inert. What it *can* do is prevent meat and milk from becoming ב"ח to begin with, because the formation of ב"ח needs to be by means of a normal cooking process (דרך בישול), and cooking with ash does not qualify as such. All of the questions are therefore resolved, because the issues under discussion in every case brought by the ט"ז and ש"ך are dealing with fully formed איסור:

- The reason the רמ"א prohibits the use of a shampoo pot for cooking food is that the ashes used for the shampoo come from beneath the stove, where hot drippings of meat and milk mingle, become ב"ח, and then get added with the ashes to the pot. Now, by the time the drippings are added to the pot along with the ashes, they have already become prohibited as ב"ח, so the ashes can't render them inert as נט"ל.
- Similarly, the soap in the מהרש"ל's case was made with חֵלֶב, which is an איסור, so the ashes don't help to make נט"ל.
- In the case of the Midianite pots as well, the issue with the pots is that they were presumably used to cook improperly slaughtered animals (נבילות), animals with terminal injuries or diseases (טריפות), or non-kosher animals (such as pigs), all of which are איסור, so the ashes wouldn't help to render them נט"ל.

Pouring Hot Water (עירוי):

If one pours water from a כלי ראשון^[1] that contains בן יומו meat flavor onto a utensil that contains בן יומו dairy flavor, the utensil that was poured on is prohibited, because we consider water poured from a כלי ראשון (referred to as עירווי מכלי ראשון) to have a similar capacity to cook as water that is still within a כלי ראשון (though not exactly the same, see below^[2]), so the meat flavor transfers from the upper utensil, through the uninterrupted stream of hot water, into the lower utensil. [The reason this is not נב"נ (Meat>Utensil>Water>Dairy Utensil) is that we treat עירווי (as far as the lower utensil is concerned, the flavor only transfers one-way) as if the stream is inside the upper utensil, and in the equivalent case where the dairy utensil is inside the meat utensil, it would be prohibited^[3] (רמ"א צה:ג) (ט"ז ס' צה ס"ק יג)].

1. see בישול - Cooking, כלי ראשון, p. ?

2. p. ?

3. see ^77f048, p. ?

The רמ"א continues that if one poured from a neutral (parve) כלי ראשון onto a meat utensil and a dairy utensil simultaneously, even if they both still have residue of meat/dairy on them, they would not be prohibited, because עירוי is not actually a כלי ראשון, כלי, we just treat it the same in certain situations. The ס"ק כ) finds the above opinion of the רמ"א difficult, because it is established law שם that עירוי cooks כדי קליפה^[1], which should mean that the meat/dairy residue should have their flavor transferred into the outer קליפה of the dairy/meat utensils, rendering that layer prohibited. The רמ"א's opinion must therefore be that although we know that עירוי cooks כדי קליפה by chicken, that's only by chicken (or similar materials) which is soft and therefore transfers easily, but for utensils, which are hard, עירוי would only be able to either absorb or impart flavor, not both, which is what is required to create a prohibition here (the water needs to absorb flavor from the meat/dairy residue and then impart that flavor to the utensil).

The ש"ך doesn't agree with this distinction, and therefore rules:

1. see Measurements, כדי קליפה, p. ?

- If both utensils have residue on them, both utensils will be prohibited קליפה כדי.
- If only one utensil has residue on it, the utensil without residue will be prohibited קליפה כדי (the residue of meat will be absorbed into the water and imparted into the dairy pot), and even the pot with the residue will be prohibited except where this will result in a major loss (הפסד מרובה) (because we are worried that the water will extract the ב"ח flavor from the now-prohibited dairy utensil and impart it into the meat utensil, although this is not a strong enough possibility to prohibit in a case of major loss).
- If they were both clean, they are permitted because any flavor transfer will be נב"נ (Meat>Utensil>Water>Utensil).
- If they were both clean, but one utensil already contained איסור, the flavor will be transferred from the איסור utensil and prohibit it, and נב"נ won't help because we don't say נב"נ by איסור.[^][There is no permit of הפסד מרובה here, unlike the dirty utensil when only one has residue, because here the עירוי only needs to absorb flavor from one place (the איסור utensil) and impart it somewhere else (the permitted utensil), as opposed to over there where the flavor needs to be absorbed from the meat residue, imparted to the dairy utensil, further absorbed from the now-ב"ח utensil, and imparted into the meat utensil.]

The Status of the Water:

Even if the utensils are not בני יומן, the wash water will still be prohibited לכתחילה, although if they are combined with food that would be a case of בדיעבד and therefore permitted. This is because ^[1]לנט"ל is אסור לכתחילה, and the only reason to permit the water is the fact that the בליעות of meat and milk it contains are not בני יומן, therefore the water is only permitted in a בדיעבד scenario. [One could ask, isn't every case where the water has absorbed these בליעות a בדיעבד scenario, the water has already absorbed the בליעות? The answer is that water alone is never considered to be a case of בדיעבד, because it's no real loss to just spill out the water and replace it (obviously in a drought or other scenario where water is scarce/valuable this would not be true), and the permit of בדיעבד only exists because "the Torah has mercy on the money of Jews" (רמ"א צה:ג). (ט"ז ס' צה ס"ק י').]

Safeguards Against Cooking:

The Stringencies of the רמ"א:

The רמ"א in פז:ו lists four stringencies that one should keep to avoid the inadvertent transgression of the laws of בשר בחלב:

1. see תערובות, נותן טעם לפגם, p. ?

1. One should not stoke the flame upon which a non-Jew's pot is cooking, because even if there is not currently meat *and* milk in the pot, the pot might have recently been used to cook meat or milk and now be cooking the opposite, which would be considered cooking בשר בחלב, due to the current meat mixing with the בליעות^[1] of the milk (or vice versa) that were previously in the pot.^[the reason this is only a stringency and not an actual prohibition is that we have a rule that סתם כלים אינם בני יומן^[2], and so the non-jews utensil presumably was not used within the last 24 hours to cook meat/milk (or anything else).]
2. One should not combine wash-water that has been used to clean meat pots with wash-water that has been used to clean dairy pots, even to give to his animal, "because such a mixture is prohibited to benefit from."^[This statement is puzzling, because, as the ש"ך points out, what would the prohibition be? The ש"ך therefore concludes that the case is where the waters were combined hot, and they are therefore true בשר בחלב. This is also problematic because, as רעק"ע points out, it would be forbidden to make such a combination to begin with, as that would be cooking בשר בחלב. Furthermore, if it becomes actual בשר בחלב, how can the רמ"א conclude that it is just a stringency not to feed the water to one's animals, it is an איסור דאורייתא of benefitting from בשר בחלב? Perhaps the רמ"א means that such a mixture would be prohibited from benefit if it were hot and so says even such a mixture when cold should be avoided, which would explain why it's just a stringency and not a prohibition.]
3. A pot used for making shampoo (the primary ingredient of which was ash at the time) should not be used to cook food, because the ashes used in the shampoo would be taken from the stove and so would contain drippings of meat and of milk from previous cooking, meaning that בב"ח of בליעות are contained in the pot.^[The reason this is just a stringency is presumably יומן סתם כלים אינם בני
4. The furnaces used to heat the houses in Europe in the רמ"א's time would have

a built-in pot for boiling water (not for food purposes). Pots of food would be placed in the furnace to cook, and the food would occasionally splash onto the walls of the built-in pot. Now, they would cook both meat and milk (not simultaneously) in these furnaces, which means that the built-in pot would contain בליעות of both. Therefore, the built-in pots should not be used for cooking food.^[This as well is presumably due to סתם כלים אינם בני יומן The כא) adds that if the built-in pot is made of metal it can be cooked in even according to the רמ"א because it's constantly being heated to the point of ליבון קל^[3] by the normal firing of the furnace.]

1. see תערוכות, בליעות & טעם כעיקר p. ?

2. see תערוכות, סתם כלים אינם בני יומן p. ?

3. see תיקון כלים, ליבון קל p. ?

Safeguards Against Combination:

Joint Placement on a Table:

It is prohibited to place meat, even of birds^[1] or undomesticated animals^[2], on a table upon which someone is eating dairy, due to the concern that the one eating meat may eat some of the other's dairy over the course of the meal or vice versa^[3] (שו"ע פח). This law was enacted against the possibility of one casually and unthinkingly eating the other's food. With that in mind, there are a few conditions for this law, without which casual and unthinking eating of the other's food would be unlikely:

1. The two people must have a basic level of familiarity with each other.. ^[The accepted opinion (שם ט"ז ס"ק ב וש"ך ס"ק ד) is that this even holds true if the two people have an animosity which would cause them not to share, because we don't want to make the law nuanced, which could lead to mistakes.]
2. The table is the one where the *eating* is taking place, this prohibition doesn't apply to a serving table.
3. The two are sitting near enough that one can reach the other's food (בשם רש"ל) (פ"ת שם ס"ק ג בשם ב"ח).
4. There is nothing placed between the two of them to serve as a reminder not to eat the other's food. ^[A third-party watcher would not fulfill this requirement (רעק"א שם), the reminder must be an object that serves as a constant reminder directly to the eaters.]

1. p. ?

2. p. 36

3. p. ?

What Qualifies as a Reminder:

As the purpose of the item being placed on the table is to be a constant reminder to those eating not to partake of the other's food, the item must be something that will continually be noticed due to the irregularity of its presence. This precludes any item that would have been on the table even had they both been eating meat or milk, such as a candelabra at night.

Separate Tablecloths:

If the two parties are eating on separate tablecloths or on separate placemats (as long as they are not accustomed to using placemats, psak from R' Tzvi Yaakov Stein), that would qualify as a reminder.

Bread:

A loaf of bread placed between the two parties qualifies as a reminder (שו"ע פח:ב) but only if the two of them are not eating from that loaf (רמ"א שם).

The רמ"א adds that although this is true, in our scenario of two people, one eating meat and the other eating dairy, they shouldn't actually eat from the same loaf because of bits of food that can adhere to the loaf.[^][This would presumably not hold true if the loaf was sliced, but I don't have any proof of this.]

A Shared Cup:

If a cup that would not normally be on the table even if it was in use, then even if they are both drinking from it it would qualify as a reminder (רמ"א שם).

The רמ"א adds that although this is true, in our scenario of two people, one eating meat and the other eating dairy, they shouldn't actually drink from the same cup because of bits of food that can adhere to the cup. Similarly, they should each have their own salt dish.^[This probably wouldn't apply if they use a covered saltshaker like is normal nowadays, as opposed to dipping plates where meat or dairy may fall in, but I again have no proof of this.]

Waiting Between Meat and Dairy:

Dairy After Meat:

One may not eat dairy immediately after eating meat, even of birds^[1] or undomesticated animals^[2] (שו"ע פט:א). This is because we are worried that the dairy will mix with the meat that was previously eaten, either because meat leaves a lingering taste in the throat when swallowed (רשי חולין ק"ה. ד"ה אסור) or because bits of meat get stuck in between the eater's teeth (רמב"ם הל' מ"א ט, כח). We rule stringently to take both opinions into account, therefore one must wait even if one chewed food without swallowing (even though, according to רשי, the waiting period is only triggered by swallowing); if one found meat between his teeth after the waiting period he must remove it[^][Finding such meat does not extend the waiting period, it continues from the time of the original eating (ס"ק ג) (ש"ך ס' פט), but the finder must cleanse his palate^[4] (רמ"א שם סע' א) and rinse his mouth (ש"ך שם ס"ק ד) before eating dairy.] (even though, according to the רמב"ם, any meat remaining between the teeth after the waiting period is no longer considered meat and need not be removed[^][This is because the רמב"ם holds that the source for the waiting period is the verse (lit. הבשר עודנו בין שיניהם) במדבר יא:לג) The meat was still between their teeth) which by, virtue of referring to particles of meat stuck between the teeth as meat, teaches us that such particles are considered meat, and even so, the רבנן said that after the waiting period, one can eat dairy, meaning that we no longer consider the particles between the teeth to be meat.]); and one must wait after eating fat like after meat[^][Even though it doesn't stick in the teeth.] (see רמ"א פט:ג).

Length of the Waiting Period:

1. p. 36

2. p. 36

3. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=105&format=pdf>

4. p. 78

The source for waiting between meat and dairy is a גמרא (ה) [1] חולין ק"ה) that prohibits eating dairy after meat in the same meal but allows dairy "in a different meal."

The שו"ע (פ"א) interprets this as requiring one to wait the standard length of time between two meals, which in the time of the גמרא was six hours^ [The hours referred to here are standard hours (60 minutes) not proportional hours (1/12 of the day or night).], before eating dairy after meat.

1. <https://www.hebrewbooks.org/shas.aspx?mesechta=31&daf=105&format=pdf>

The רמ"א (שם) brings another opinion that interprets the גמרא completely differently. According to this interpretation, when the גמרא prohibits dairy after meat until "the next meal" it means that literally, so that all one would need to do is finish the current meal, say a ברכה אחרונה¹ [The רמ"א brings an opinion that one shouldn't make a ברכה אחרונה during his meal in order to eat dairy, but rules that we don't need to follow this opinion. The רמ"א (שם ט"ז) vehemently argues on this רמ"א because the thing that makes one allowed to eat dairy is the fact that it's being eaten during a separate meal. It can't be considered a separate meal if one is eating continuously, stopping only to make a ברכה אחרונה. The ברכה אחרונה would only help if the first meal had been legitimately concluded (such as by davening mincha, like many do on shabbos to eat shaleshudos). The רמ"א goes so far as to say that "one who fears the word of G-d should avoid sitting at the same meal with those who are not scrupulous in this matter"], and start a new meal of dairy.² [According to this opinion, if he didn't make a ברכה אחרונה, then so long as the meal during which he ate meat can still be considered ongoing, he cannot eat dairy, even after six hours pass. (ש"ך ס' פט ס"ק ה)] According to this opinion he would also need to cleanse his palate^[1] and rinse his mouth[^] [The order is not important, rinsing first would also be effective (ש"ך שם ס"ק טו).] before eating dairy.^{806c6e}

The רמ"א then brings the custom of his region which was to wait an hour between meat and dairy as a sort of compromise position between the two interpretations. Those following this custom are required to adhere to the stringencies of the רמ"א's first opinion^[2], namely to make a ברכה אחרונה and cleanse the palate.

1. p. 78

2. p. ?

The רמ"א ends off that there are those that are "fastidious" to wait six hours, and that this is the proper thing to do.

If one just chewed meat without swallowing, there's no need to be stringent and wait six hours instead of one. This is because according to רש"י^[1] there is *no* waiting period required in such a situation, as the lingering flavor he considers the basis of the waiting requirement lingers in the *throat* and is therefore only triggered by *swallowing* meat, so in this case one can rely on the combination of רש"י who would require no waiting and those who generally only require an hour (רעק"א שם). Presumably he would apply this same logic to one who ate fat, because fat doesn't stick in the teeth and so would not trigger any waiting period according to the רמב"ם.

Cleansing the Palate (קינוח):

In order to remove flavor from one's mouth, one must do קינוח (lit. wiping), which we will refer to as cleansing the palate. It is accomplished by eating something that will scrape his mouth somewhat, like bread or a cracker, but not with something sticky, like flour, dates, or vegetables, as they don't work well for this purpose (שו"ע פט:ב). The food must be swallowed, not just chewed (שם ס"ק ה). (פ"ת).

Meat After Dairy:

1. p. ?

Unlike eating dairy after meat^[1], there is no waiting period to eat meat after eating dairy (שו"ע פט"ב). This is because the two reasons that are given for the post-meat waiting period, namely lingering flavor and pieces stuck between the teeth, do not apply to dairy, which doesn't stick in the teeth or leave a lingering flavor. Therefore, one who eats dairy and wants to eat meat need only check his hands for dairy residue[^][The שם ס"ק ה) and ר"פ and פמ"ג (הובא בפ"ת שם ס"ק ד) (The שם ס"ק ה) (פ"ח הובא בב"ה הנ"ל)] say that unless one doesn't have water, he should be stringent to wash his hands even by day when there is ample light, unless he is eating with utensils, in which case he doesn't need to wash his hands (שם ס"ק ט) (or, at night when there's not enough light[^][If one had access to a good light (such as modern electric lights) then he can just check his hands, like by day (שם ס"ק ט) (ש"ך הובא בב"ה שם ס"ק ו) (ש"ך)], wash them[^][With water, not other liquids (ש"ך הובא בב"ה שם ס"ק ו) (ש"ך)], cleanse his palate^[2], and rinse his mouth. Even this is only necessary for the meat of a domesticated[^][because it is included in מדארייתא בב"ה so we treat it more stringently.] or non-domesticated[^][because it is visually similar to the meat of a domesticated animal, even though its only included in (ש"ך שם ס"ק יד) (מדרבנן) animal, bird meat can be eaten immediately[^][because it is only included in בב"ה and is not very similar to a domesticated animal.].

The שם (רמ"א) cites a stringent opinion that requires waiting for the same period as that of meat to dairy^[3] even from dairy to meat (even the meat of birds). He then rules that one should be stringent like this opinion when eating hard cheese^[4], because it can get stuck between the teeth and it has a strong flavor that lingers, but there are opinions who are lenient and so we should not protest those who don't wait even after hard cheese.

1. p. 74

2. p. 78

3. p. 75

4. p. 80

The (הוא) describes the terrible spiritual harms that befall one who eats meat and milk "together, in the same hour^ [although the word used can be translated as "in the same time period", implying less than an hour], or in the same meal." Because of this, there are those who are stringent to wait an hour between dairy and meat. Those who wait half an hour are presumably combining this with the principle of רובו ככולו (a majority is akin to the whole) to say that waiting just over half an hour is effectively the same as waiting an hour.

The (פסד) continues that in order to eat meat after dairy on the same table where he ate the dairy (or vice versa^ [this was presumably discussed in a case of meat after dairy because there is no/a limited waiting period^[1] after dairy, so it's more likely that one will want to use the same table before it has been cleaned.], see (רמ"א שם) he must clear away all of the remnants of bread on table and must change the tablecloth. The (שם ס"ק ח) says that nowadays that we eat from plates, rather than directly off of the tablecloth, we can flip the tablecloth rather than change it.

What Qualifies as Hard Cheese:

The (שם) ש"ך says that any cheese that has been aged six months is considered hard. This is the generally accepted practice of those who wait after hard cheese.

The (שם) ט"ז says that only wormy cheese (such as Casu Martzu^[2]) or cheese that has been curdled with rennet^[3] (see (פ"מ מ"ז שם ס"ק ד) has the lingering taste required to be considered hard cheese.

Do Waiters Need to Wash Their Hands:

1. p. 78
2. https://en.wikipedia.org/wiki/Casu_martzu
3. p. 40

The (פס"ג) רמ"א says, based on a רש"י, that the requirement to wash one's hands between meat and dairy only applies to those who are eating, not waiters who just serve the food.

The (שם) ש"ך vehemently argues on the רמ"א and says that there is no difference between the waiters and the diners, and so waiters that touch meat must wash their hands before touching dairy and vice versa, and that the רש"י cited by the רמ"א is saying that the waiter doesn't need to wash their hands before handling bread, nothing to do with בב"ח.

(תבשיל של בשר או חלב): Dishes Containing Meat or Dairy

If one eats a dish that contains meat (such as cholent) but doesn't eat any of the meat itself (he ate only the potatoes¹[If he rinsed them, otherwise he must assume that there was liquefied fat present on the potato.]), he can immediately eat an equivalent dairy dish (Ex. the onions out of a French onion soup), or vice versa, with no need to wash his hands (although it is recommended) (שו"ע פס"ג). The (שם) רמ"א says that one must wash his hands between.

If after eating such a meat dish one wants to eat actual dairy, or vice versa, he must wash his hands in between (שו"ע שם). The (שם) רמ"א says that the custom has become not to eat actual dairy even after a meat dish just like after full-fledged meat^[1]. (שם) רעק"א says that by his time the custom had become to wait after a meat dish to eat a dairy dish as if between meat and dairy.

1. p. 74

The רמ"א continues that after a dairy dish one would be permitted to eat meat right away, but he must wash his hands, even from a dairy dish to a meat dish. One would also be permitted to eat dairy immediately after a parve (neither meat nor dairy) dish cooked in a pot that had previously been used for meat^[Even if the pot was not cleaned well. We can derive this from the fact that the רמ"א permits eating dairy immediately *after* the dish, not *with* the dish, which means there must be at least a small amount of meat present, otherwise the dish should be able to be eaten directly with the cheese (ש"ך שם ס"ק יט).].

Utensils of Meat and Dairy:

One cannot use a knife that is set aside for meat to cut dairy (שו"ע פט:ד), or vice versa (רמ"א שם). In fact, it is even prohibited to cut *bread* that is to be eaten with dairy/meat using a meat/dairy knife. This prohibition is only against initially cutting dairy with a meat knife, but if one already cut the dairy it is not prohibited unless the knife had been used to cut boiling hot meat^[1] within the last 24 hours^[2] and the dairy is itself boiling hot.

Removing Meat or Dairy Status From a Utensil:

1. see תערובות, יד סולדת, p. ?

2. see תערובות, בן יומו, p. ?

טבילת כלים:

Metal Knives:

Metal knives have an extra method of having their meat/dairy status removed, and that is to stab them repeatedly into hard earth (נעיצה בקרקע). That said, this method is not ideal, and the custom has become to have separate knives for dairy and meat, respectively, and to mark[^][These marks were seemingly things like notches or other marks that are not clear in and of themselves that they were marking the knife as dairy, which is why the רמ"א says not to differ from the custom to mark the dairy knives specifically, as marking the meat knives instead would lead others assume that the meat knives were, in fact, dairy.] the dairy knives[^][Nowadays that we use stickers that say חלבי and בשרי or knives with blue and red handles, it seems that there would be no reason to mark the dairy knives specifically, as our method of marking is unambiguous and a meat knife so marked will be obviously meat to an observer.], and one should not differ from the custom[^][According to this, it would presumably be, at least, discouraged nowadays to have a red knife for dairy or a blue knife for meat, as any outsider using the kitchen would think the red is meat and the blue is dairy.] (רמ"א פט"ד:). The שם (ש"ך) adds that although this is true, in an extenuating circumstance wherein one only has access to one knife he may perform נעיצה to switch between meat and dairy.

Salt Dishes:

It is prohibited to place a salt dish (an uncovered utensil containing salt) near a meat or dairy dip, lest some of the dip fall into the dish, get hidden by the salt, and then, when the salt is used at a different meal of the opposite kind (milk or meat), the meat/milk dip that fell into the salt will get combined, along with the salt, with the milk/meat of the meal (שו"ע צה:ה). This stringency would not apply to a bowl of vinegar, even though it is also placed on the table at meals (שם), because the dip will dissolve in the vinegar, and therefore has the ability to become בטל, so we're not as worried (ט"ז וש"ך שם).

This law only applies if the salt dish is uncovered, as there is a possibility the dip will fall in, and is only לכתחילה. If one did place the salt near the dip, we do not need to assume that the dip fell into the salt (רמ"א שם).

Salt in Meat/Dairy Utensils:

If one uses a clean, בן יומי meat/dairy utensil as a salt dish, the salt does not absorb flavor of meat or milk and can therefore be used for dairy/meat (שו"ע צה:ז). This would also be true for a clean, בן יומי utensil of איסור, the salt would not absorb איסור flavor and would remain permitted for use with other food (ס"ק כה) (ש"ך שם).

The רמ"א rules that there are those who are stringent not to place salt in a meat/dairy dish[^][Specifically if the utensil is made of earthenware (כלי חרס) (צה"ק ח) if he might use it later with dairy/meat, and one who is stringent like this opinion "תבא עליו ברכה".

Contact:

The combination that we are safeguarding against is one that cannot be undone, such as liquids mixing together or בליעות of milk penetrating meat (or vice versa). Contact between cold solids (meat and cheese), however, is not something we are very concerned about, as all that needs to be done to rectify the situation is to rinse^[1] (הדחה^[2]) the points of contact between the meat and cheese^[3].

Additionally, the transgression if one were to eat it without הדחה would only be rabbinic in nature^[4]. Furthermore, if one were to roast or salt the meat or cheese so that the residue is absorbed כדי קליפה^[5] (and certainly if one were to cook the meat so that the residue is distributed throughout the entire piece of meat), the residue is such an insignificant amount that the meat or cheese wouldn't even need קליפה (see פ"ת ס' צא ס"ק א). Therefore, one can even wrap meat and cheese in the same bundle and not worry that they might come into contact with each other (שו"ע צא:א).

Bread:

1. p. 30

2. see Measurements, Surface Level, p. ?

3. p. 30

4. p. ?

5. see Measurements, כדי קליפה, p. ?

One must be careful not to allow his bread to come into contact with meat or dairy (other than bread he is currently eating with the meat/dairy), lest it come into contact with meat and subsequently be eaten with dairy (or vice versa) (צא:ג (שו"ע). If the bread and meat *did* come into contact, the bread requires הדחה^[1] (see ש"ך שם ס"ק ד if the meat touched the crust and גרידה^[2] if it touched the crumb (ד (פ"ת שם ס"ק).[^][This differs from the normal law regarding rinsable residue, לכתחילה, where, while in most cases we can't allow the items to come into contact, we do not need to be careful to avoid contact. The reason bread is different is that bread is commonly eaten with both meat and cheese, and so there is a strong likelihood that a piece of bread that touches meat will be eaten with cheese and vice versa (פ"ת שם).]

Storage:

One is allowed to store milk and meat near each other in a closet (or in a refrigerator) (שו"ע צה:ו), because people are careful to keep meat and milk from mixing (ט"ז שם ס"ק יז). The רמ"א (שם) says that although this is true, one should be stringent to keep them separate if he doesn't have a specific need (such as storage constraints). The ש"ך (שם ס"ק כד) says that this stringency only applies when the meat and milk are uncovered, if they're covered there's no concern that they will mix.

1. see Measurements, Surface Level, p. ?

2. see Measurements, גרידה, p. ?

Meliha — Salting

מליח כרותח

Salted food is capable of emitting its בליעות^[1] into food it comes into contact with. This law is referred to as מליח כרותח, which directly translates as "Salted is like Heated", because the primary method of transferring בליעות is heat. This only applies to meat that is too salty to be eaten as is (אינו נאכל מחמת מלחו).

1. see תערובות, בליעות & טעם כעיקר p. ?

Levels of Salting:

Not just any salting qualifies as מליח כרותח; there are specific requirements for both the quantity of salt required, as well as for how long the salt must be on the food.

מליח לקדירה:

The most basic level is מליח לקדירה (lit. salting for the pot), which is the level of salting required to kasher (draw out the blood) meat, which requires fully coating the surface of the meat with salt. The status of מליח כרותח begins immediately and continues for the length of time it takes a pot to boil, which is approximately 18 minutes, at which point all of the blood has been expelled and the salt's strength is considered spent.

If this level of salting was applied to kashered (bloodless) meat (or any other substance), the status of מליח כרותח doesn't start until after 18 minutes, and it continues so long as the salt is present, because there's no blood extraction to spend its strength (שו"ע צא:ה).

The reason for this distinction is that salt is not רותח in a vacuum — it needs to be acting on something. Therefore, when bloody meat is salted and immediately begins expelling blood, it is immediately considered רותח. However, when the meat doesn't contain any blood, nothing is happening when the salt is first placed on the meat and so there is no רתיחות. Once the salt has been on for 18 minutes, the meat itself absorbs the salt, and salty meat is always considered רותח.

The שם ס"ק יא) says that this distinction as far as when the רתיחות starts is the opinion of the רן and is rejected by nearly all other ראשונים. He therefore rules that one must treat even bloodless meat as רותח as soon as the salt is placed upon it.

The (רמ"א) (צ"ח) cites an opinion that disagrees with this distinction as far as when the *ends* and says that even bloodless meat ceases to be *רותח* after 18 minutes. He rules that one may rely on this opinion in extenuating circumstances, such as a significant financial loss (הפסד מרובה) or if the food is needed for a *מצוה*.

מליח לצלי:

The lowest level of salting is *מליח לצלי*, which is the amount of salt that one places on meat to season it for roasting (or any other cooking method). This level of salting has no effect as far as *מליח כרותח*, and if meat that is salted to this level touches cheese, it is treated the same as if it was unsalted^[1]. The (רמ"א) (צ"ח) says that nowadays that we are not experts in what is considered *מלחו* *מחמת*, even *מליח לצלי* should be treated as *רותח* except where such a ruling would result in a significant financial loss.

Drippings:

Although meat that is *מליח לצלי* is itself not *מלחו מחמת* and is therefore not *רותח*, the drippings of such meat is *מלחו מחמת* and, therefore, *רותח* (see (רמ"א שם)).

מליח להצניע:

The level above *מליח לקדירה* is *מליח להצניע*, which is salting to preserve for storage.

מליח לדרך:

1. see בשר בחלב, General Contact, p. ?

The most intense level of salting is מליח לדרך, salting to preserve for travel, which needs a higher level of salting than salting for storage^[1] so that it can withstand exposure to the elements.

1. p. 105

How "מליח כרותח" is "רותח"?

When food is מליח כרותח and therefore obtains the status of מליח כרותח, what flavor transfer(s) this effects is subject to dispute. The רמ"ע (צא:ה) rules that salted food imparts flavor but does not absorb flavor, and it imparts its flavor to the depth of a קליפה^[1] (when it's not fatty^[2]). Therefore, if salted meat comes into contact with unsalted cheese or vice versa, the unsalted food needs קליפה while the salted food just needs הדחה^[3]. If both were salted then both need קליפה, as each imparts its flavor to the other. If both items were dry they don't impart flavor, even though they are salted. If even one of them is moist, even only the unsalted, then the salted imparts its flavor using the moisture as a medium. The רמ"א brings an opinion that flavor transfers both ways, even when only one is salted and, although we don't rule like that opinion, says we should be stringent when doing so will not lead to a financial loss (קה:י). The רמ"א (שם) rules that this is when the salted only imparts and doesn't absorb is when the unsalted substance is solid; If it's liquid, the salted substance will absorb some of it[^][If the liquid was not diluted by water. If it was diluted, the water within it is considered to have rinsed the meat and removes its status of מליח כרותח to the point that it won't absorb the liquid.]. If the salted substance was fully dry, however, it wouldn't even absorb liquid[^][This obviously only holds true until the liquid penetrates the dry substance and rehydrates it.].

Depth of Flavor Transfer:

1. see Measurements, כדי קליפה, p. ?

2. p. 91

3. see Measurements, Surface Level, p. ?

Food that reaches the status of מליח כרותח imparts its flavor to the depth of a קליפה, unless the salted food is fatty, in which case the fat carries the flavor throughout the second substance. Not only that, if the second (non-fatty) food is also salted and would normally only impart its flavor to the depth of a קליפה, the fat of the first food permeates it and makes the second food fatty as well, which in turn imparts its flavor throughout the first food (שו"ע צא:ו).

The רמ"א (קה:ט) says that we are not experts at discerning between fatty and not fatty and so anything that can be fatty, like meat, must be treated as such.¹ [This is the second מליח scenario in which the רמ"א says to be stringent because we don't have the requisite expertise to make the distinction require to be lenient, the first being מליח לצלי.¹] רעק"א says (יו"ד ס' צא ד"ה וטוב להחמיר) that if one had a non-fatty food that was only salted לצלי, we won't doubly make the argument of "we are not experts", and so the flavor will only transfer כדי קליפה, although he brings a פ"מ that argues.]. We don't have such a concern about substances that *can't* be fatty, such as חמץ or utensils, and so they only impart flavor to the depth of a קליפה.

1. p. 105

Taarovot — Mixtures

ביטול:

Under basic Torah law, in any mixture, so long as it is not perfectly even, the minority is nullified in the majority and the mixture is treated as purely consisting of the majority. The primary consequence of this is that if something that is forbidden to be eaten gets mixed with food that is permitted (in a scenario where the two components are visually indistinguishable), so long as the permitted food makes up more than 50% of the mixture, the entire mixture may be consumed, with no concern for the prohibited component.

There are, however, myriad exceptions and qualifications to this rule:

בליעות & טעם כעיקר:

When food is cooked, it imparts flavor into the utensil it is being cooked in and to any other food or liquid cooked with it. This flavor is called בליעות (a plural term that literally translates as "absorptions", referring to the particles of flavor that have been absorbed). When the utensil is later cooked in, the בליעות are then emitted by the utensil into the new food being cooked. Now, the בליעות emitted by the utensil shouldn't be relevant, as they will be the minority component of any mixture and will therefore be nullified. There is, however, an exception to the rule of ביטול called טעם כעיקר (lit. flavor akin to essence), which means that so long as something can be tasted in a mixture, even if there is only flavor and no physical essence present, it is considered to be present and not nullified. If איסור was mixed with היתר, the איסור שו"ע (צב:א) holds that the mixture can be given to a non-jew to taste, and if he doesn't detect the taste of the איסור in the mixture, it's permitted. The רמ"א argues, and says that we don't rely on a non-jew. He therefore rules that we must calculate if the איסור is בטל בששים^[1].

ביטול בששים:

The רבנן calculated that the ratio required to nullify flavor is 60:1 (60 parts permitted substance to one part flavor). At this ratio the flavor is assumed not to be present. This is referred to as ביטול בששים (lit. Nullification in 60).

מין במינו:

Until now we've been discussing mixtures of substances with differing flavors, which makes ביטול simple: when the minority flavor can no longer be detected, there's no more טעם כעיקר. However, when the two substances have the same flavor (such as permissible milk mixed with prohibited milk), there are two, opposite approaches:

1. p. 105

- According to רש"י, the prohibited milk is never בטל, because the mixture always tastes like milk, and the prohibited milk is contributing to that taste, no matter how little.
- The accepted opinion is that of ר"ת, who holds that the milk is בטל at the same ratio of 1/60 as any other substance, there's just no recourse of having a non-jew taste the food, even according to those that allow that by differing substances.

קליפה:

If a prohibited substance is absorbed into a permitted substance כדי קליפה^[1], the קליפה must be removed. If, however, the קליפה is not removed and the permitted substance is cooked, the cooking distributes the prohibited substance throughout the permitted substance, and it is therefore בטל^א [assuming the קליפה is less than 1/60 of the total, which it almost certainly is] (שו"ע יו"ד צא:ד והגה שם), although the קליפה itself must still be removed. If the permitted substance was a liquid, there is no קליפה to remove because the liquid that touched the איסור does not stay put. Therefore, so long as the liquid has 60 times the volume of what would be a קליפה, it's מותר (see ט"ז שם ס"ק ז).

Exceptions to ביטול בששים:

There are a few things that the רבנן said are not בטל בששים and in fact are never nullified:

מעמיד:

Something that gives the food its structure is not nullified no matter how little is present. This is because even if there is barely any of the prohibited substance present, its presence is felt because it gave the food its structure.

1. see Measurements, כדי קליפה, p. ?

This is only true if the מעמיד is intrinsically prohibited, but if it's only the combination of the מעמיד with the primary substance that is prohibited than the מעמיד would be nullified. (רמ"א יו"ד פז:יא) The רמ"א adds that even if the מעמיד is intrinsically prohibited it would still only not be nullified if there wasn't another, permitted מעמיד present. If there was another such מעמיד present, we would say זה וזה גורם (lit. this and this caused) and the mixture would be permitted.

The רמ"א (that the רמ"א adds a qualification to this ruling of the רמ"א (שם ס"ק לו) seems to disagree with) that we only say זה וזה גורם if there wasn't enough of the prohibited מעמיד to be effective on its own. If there was enough of the prohibited מעמיד, the mixture would be prohibited regardless of the permitted מעמיד.

יד סולדת

The minimum heat required for the transfer of בליעות, (except in the case of כבוש^[1], cutting sharp food with a knife, מלוח, or possibly cold smoking^[2]) is יד סולדת (lit. the hand recoils), the temperature at which a person touching the heated item would recoil due to the heat. There are various opinions as to what this temperature is, ranging from about ~104°F to ~165°F.

נותן טעם לפגם

The exception to טעם כעיקר is if the flavor being imparted is detrimental to the dish it enters, this is called נותן טעם לפגם (lit. imparts flavor deleteriously). It is important to note that נותן טעם לפגם is only permitted בדיעבד, one is not allowed to cause food to absorb איסור that is נותן טעם לפגם intentionally.

1. see בשר בחלב, Soaking (כבוש), p. ?

2. see 237, בשר בחלב, p. ?

There are three kinds of נותן טעם לפגם: things that are inherently disgusting, things that are not inherently disgusting but produce a bad flavor when combined with specific items, and בליעות that have been absorbed in a utensil for 24 hours.

בן יומו:

When בליעות have been contained in a utensil for 24 continuous hours they are considered to be disgusting and are נותן טעם לפגם. Up until 24 hours בליעות are referred to as בני יומן (of their day) and the utensil containing them is referred to as בן יומו (of its day), after 24 hours they are not בני יומן and are therefore irrelevant except in the case of דבר חריף.

סתם כלים אינם בני יומן

We have a rule that a utensil that we don't know how long it's been since it was cooked in is assumed to be אינו בן יומו, and therefore, in a normal case, its בליעות cannot affect anything cooked in it.

חענ"נ:

חתיכה עצמה נעשית נבילה (חענ"נ)

Egg Shells:

Egg shells are porous, therefore eggs cooked, even in their shells, absorb flavor from, and impart flavor to, what they cooked with. Therefore, eggs of איסור will prohibit foods they are cooked with and eggs cooked with meat, dairy, or איסור will take on those attributes, becoming prohibited with dairy, meat, or completely, respectively (שו"ע יו"ד צה:ב).

Sharp Foods (דבר חריף):

Sharp foods, either intrinsically sharp foods like radish and beets or mixtures consisting of at least a majority sharp foods (רמ"א יו"ד צה:ב), have special laws when it comes to the transfer of בליעות:

When Sliced With a Knife:

Slicing sharp foods, such as radishes, with a knife (even if it's cold and clean) causes the knife to expel the בליעות that it contains כדי נטילה^[1] from the place of contact with the knife into the radish (שו"ע יו"ד צו:א). This expulsion of בליעות, which normally requires heat, is effected by the combination of the pressure exerted by the knife against the food and the sharp flavor of the food (שם ס"ק ב). Consequently, if the knife contained flavor of איסור, the radish will need to have כדי נטילה removed along the place of contact before it can be eaten. If the radish was cooked in a dish without first removing the נטילה, thereby spreading the בליעות of איסור throughout the dish, the dish will need to be 60 times larger than the נטילה in order to be permitted.

If the radish was diced (so that each piece is no bigger than a נטילה) with a knife containing איסור, then if the radish was cooked in a dish, the dish would need to be 60 times the entire radish in order to nullify the איסור (even if the knife blade is smaller than the radish, meaning it didn't have enough בליעות to fully permeate the radish, we need to be worried that enough איסור entered each piece of the radish to render the piece אסור, and then חזק"נ^[2] would require you to consider each piece fully and intrinsically אסור).

Reinvigorating the flavor (מכליא לשבח):

1. see Measurements, כדי נטילה, p. ?

2. see בשר בחלב, חתיכה עצמה נעשית נבילה (חזק"נ) p. ?

When a sharp food is involved in a transfer of בליעות, such as when it is sliced with a knife or cooked in a pot which contain בליעות of איסור, even if the בליעות are more than 24 hours old and therefore לנט"ל^[1], the sharpness of the food reinvigorates the בליעות, rendering the sharp food אסור.

There is a dispute whether this applies to all sharp foods or just חלתית, which is an exceptionally sharp food and is the only food about which this law is stated explicitly in the Gemara.

Residue:

When solid היתר comes into contact with solid איסור, in order to consume the היתר one must first rinse^[2] the point where the היתר touched the איסור, because we assume that the איסור left residue on the היתר (see שו"ע יו"ד צא:א).

Therefore, one may not enter any situation that necessitates הדחה deliberately, lest he forget to rinse the food before eating. That said, in a situation where the food is usually rinsed before consuming regardless, such as a piece of raw meat, one may deliberately put it into a situation that would necessitate הדחה^[3] רעק"א brings a ראה, and the פ"ת brings the פמ"ג, that this only applies to a food item and a utensil, but it would be prohibited to bring two food items (one of איסור and one of היתר) into contact לכתחילה, although the רשב"א, and the שך, argues and permits it unless the איסור is fatty, and therefore its residue would require שפשוף^[3], because there is no real concern that it won't be rinsed (שו"ע צא:ב).

1. p. 105

2. see Measurements, Surface Level, p. ?

3. see Measurements, Surface Level, p. ?

One only needs to perform הדחקה in a situation where at least one of the items coming into contact with each other are moist so it leaves residue on the other¹[I am unsure whether one would need to rinse just the dry one, because it has residue on it from the moist one, or both because we are concerned that the moisture of the moist one draws residue from the dry one. I am leaning towards the latter option.], but if they are both dry then their touching has no consequence (ש"ך ס' צא ס"ק א).

Utensils:

If one places cold food of היתר in a cold utensil of איסור, the שו"ע (יו"ד צא:ב) says that the food must be rinsed before consumption. The רמא (שם) adds that הדחקה would only be required if the food was moist, and that if the utensil had only absorbed איסור through a cold transfer¹ [כבוש or מלוח] and the food was dry, it does not need הדחקה and may even be placed in the utensil לכתחילה.

The איסור (שם ס"ק ג) emends the רמא to say "if the utensil had only absorbed איסור through a cold transfer **or** the food was dry" because either condition is enough that there is no risk of flavor transfer. He also adds that the utensil must be clean, because if it's not then there's no scenario in which the food would not need הדחקה.

The ש"ך (שם ס"ק ג) forcefully rejects the explanation of the ט"ז, because if the utensil is clean, what transfer of residue or flavor is there for הדחקה to remove? Additionally, the ש"ך is baffled by the רמא's distinction between the utensil absorbing flavor through a heat-based transfer or a cold-based transfer. How does the way the flavor was absorbed by the utensil have any affect on whether it necessitates הדחקה?

1. see [בשר בחלב, Soaking (כבוש), p. ?]

The ש"ך therefore argues on the רמ"א and ט"ז and rules that הדחקה is only necessary if the utensil was unwashed. If the utensil is clean, then there is no reason to require הדחקה and one can even place the food in the utensil לכתחילה, although only if this is a temporary arrangement, otherwise we are worried he might forget and use the utensils for hot food (which would transfer בליעות of איסור into the food). For example, it would be permitted to use a hotel's bowl and spoon for cereal, because one is only staying at the hotel for a short time, but if one were to sign a long-term lease on a fully furnished apartment formerly occupied by a non-jew, he would be required to Kasher^[1] the utensils before using them.

1. see תיקון כלים, הסרת בליעות 1.

Tikun Keilim — Vessel Rectification

There are two main scenarios that utensils would need rectification to be usable with food:

טבילת כלים:

הסרת בליעות:

When utensils absorb undesirable בליעות (whether non-kosher, חמץ before פסח, or meat/milk when the owner wants to use the utensil for milk/meat), the method for removing those בליעות depends on two factors: the material from which the utensil is made and the method of cooking that imparted the בליעות.

Materials:

Earthenware:

Earthenware can never be fully purged of בליעות, so an earthenware utensil that absorbs from forbidden foods will forever prohibit anything cooked in it. If it absorbed from חמץ it cannot be used on פסח and in fact must be destroyed before פסח if not sold to a non-jew, and if it was used for meat/milk it can never be used for milk/meat.

This is all the case if the flavor was absorbed throughout the utensil. If, however, it was only absorbed to the depth of a קליפה^[1], one has the option to remove the קליפה (through sanding, for example) if he so desires (ש"ך יו"ד ס' צא ס"ק יח), although the (שם ס"ק ח) discusses the possibility that earthenware always absorbs throughout which would render קליפה irrelevant.

Glass:

According to most authorities, glass is incapable of absorbing בליעות and so can never become prohibited (or חמץ or meat/milk) to begin with.

Metal:

1. see Measurements, כדי קליפה, p. ?

Metal utensils can be purged of בליעות but the method required depends on how the בליעות entered. The removal method needs to be at least as intense as the way the בליעות entered to begin with, therefore בליעות that require a lower method can be purged by a higher method. The methods are:

הגעלה:

Utensils that have absorbed בליעות through cooking require הגעלה, immersion in boiling water.

ליבון קל:

Utensils that require הגעלה^[1] can, in situations where הגעלה isn't practical, be heated to יד סולדת^[2] and that suffices to remove the בליעות.

ליבון גמור:

When a utensil has absorbed בליעות through exposure to an open flame, such as roasting or grilling, it requires ליבון (lit. whitening), which is heating the metal to the point that it, according to various opinions, is red hot, emits sparks, or achieves the temperature the food was roasted at.

Wood:

Wood seems to be treated like metal^[3], although ליבון גמור^[4] is obviously not applicable, because wood can't be heated to that temperature without burning. Like earthenware^[5], one also has the option to be קולף the wood if it absorbed something כדי קליפה.

1. p. ?

2. see תערוכות, 764657, p. ?

3. p. 104

4. see תיקון כלים, ליבון גמור, p. ?

5. p. ?

General Laws:

Drippings:

If hot milk or fat falls on an unknown minority of a utensil, the utensil can be treated as neutral (not meat or dairy) and permitted because the dairy/meat/prohibited section, being that we don't know where it is and it's a minority, is בטל ברוב, although if it's a material where ^[1]הגעלה is an option, הגעלה should be performed (פ"ת יו"ד ס' צא ס"ק ח).

1. p. {PAGE}